

Tomoka Marsh Aquatic Preserve

SEACAR Water Quality Analysis

Last compiled on 15 July, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

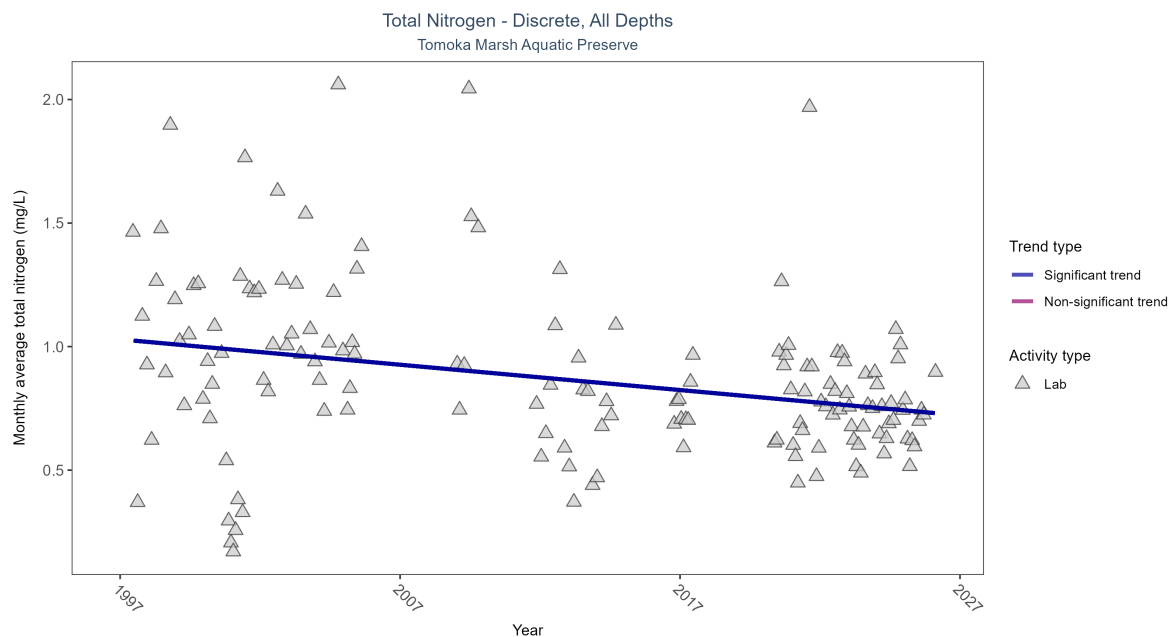


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Monthly average total nitrogen decreased by 0.01 mg/L per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	426	23	1997 - 2026	0.7824	-0.23326	1.02911	-0.01025	0.0001

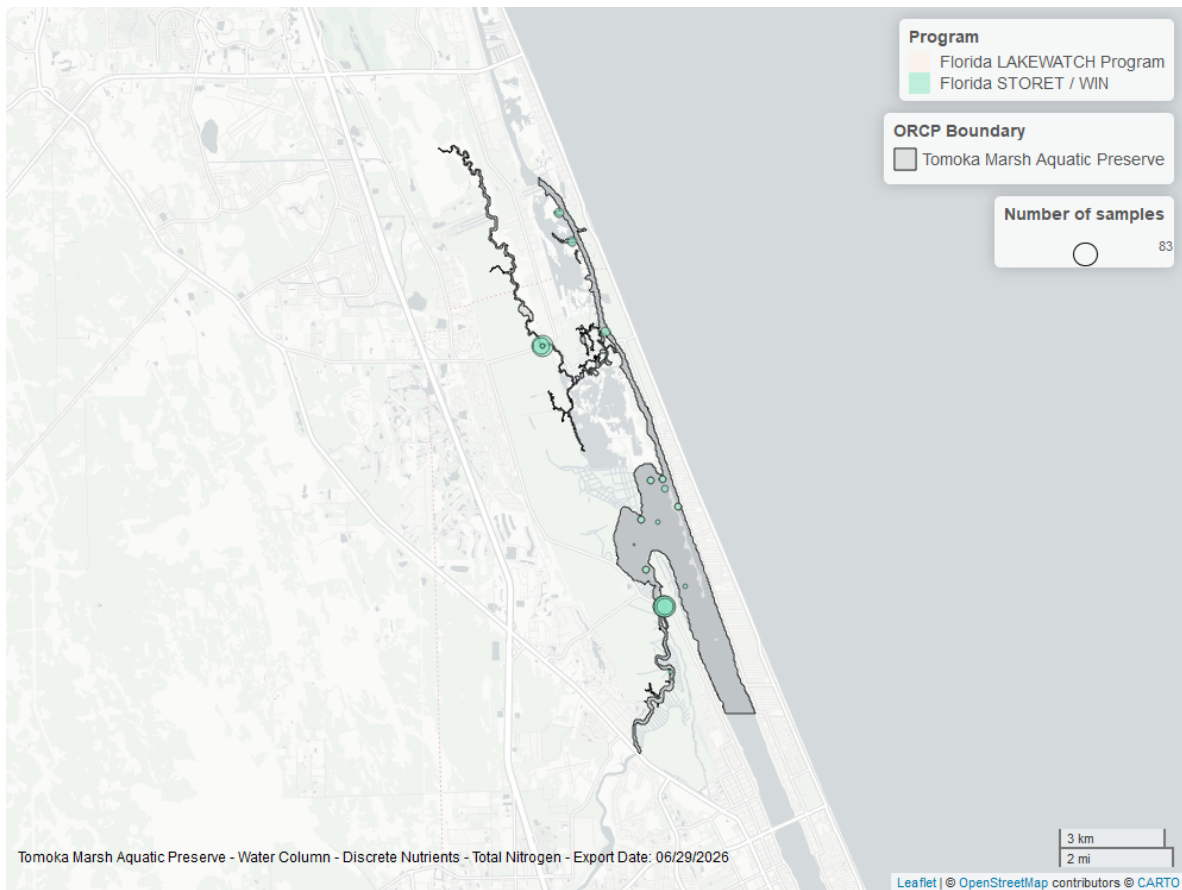


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

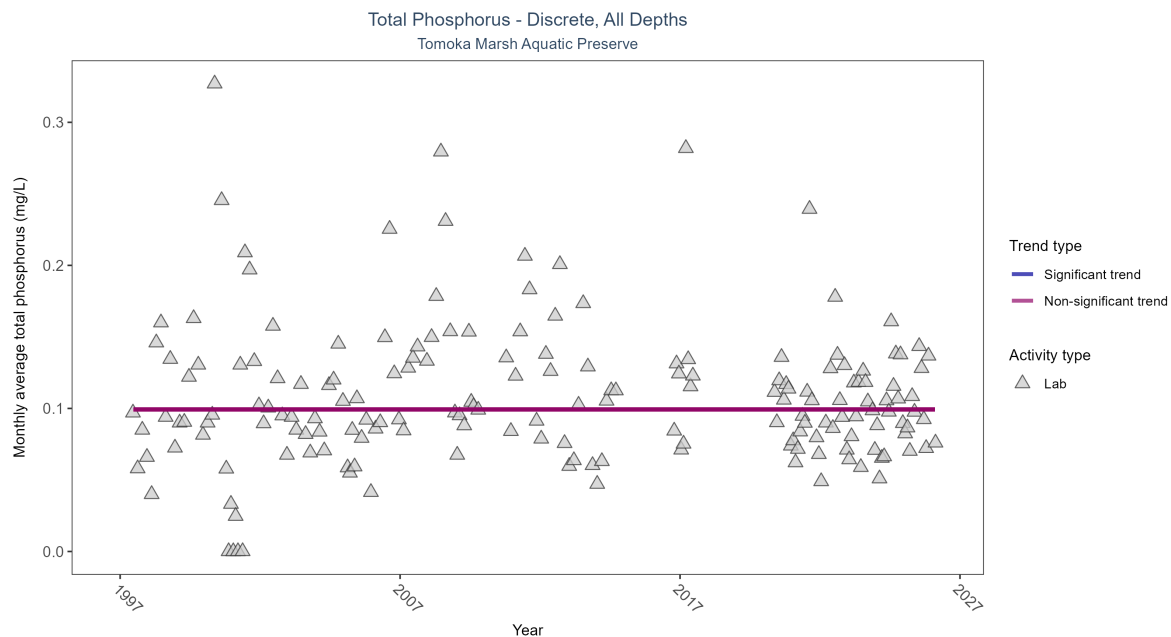


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Total phosphorus showed no detectable trend between 1997 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	799	27	1997 - 2026	0.0968	0.02288	0.09927	0	1

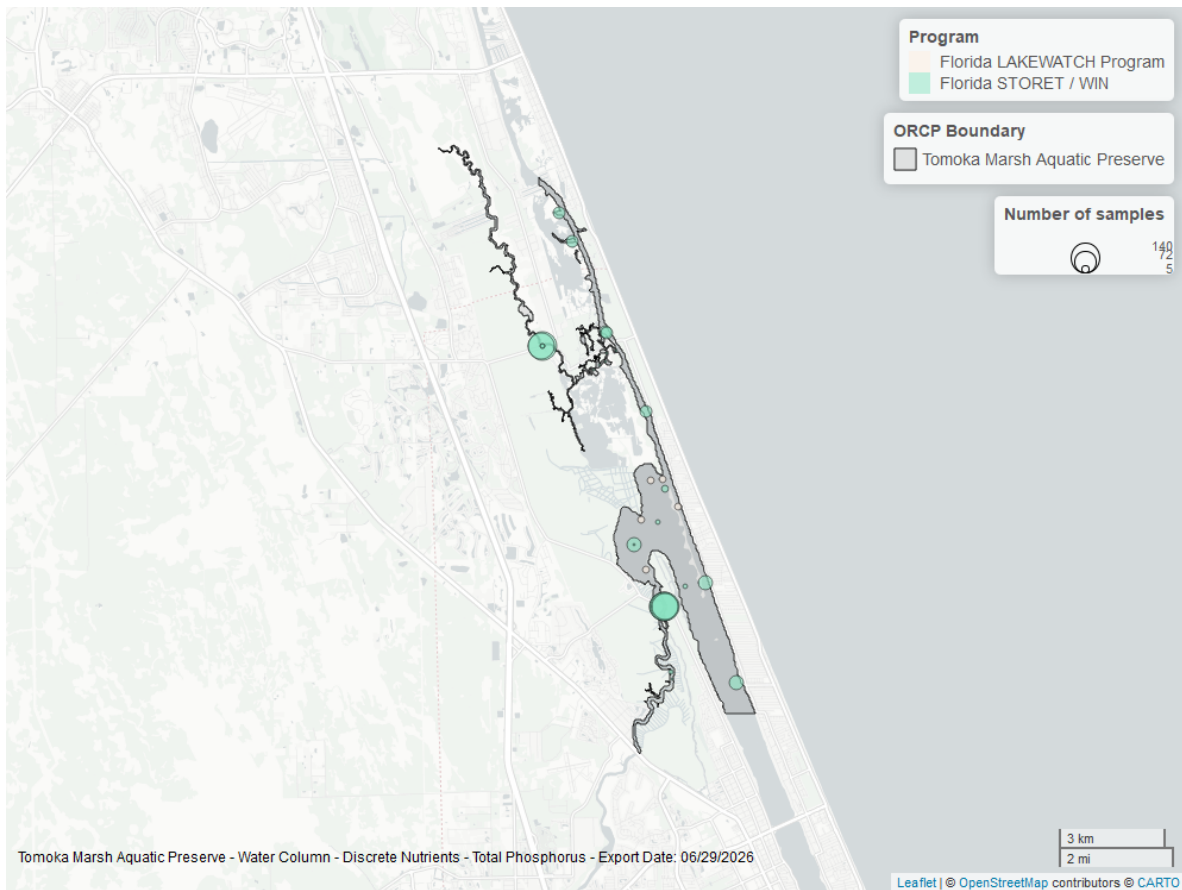


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

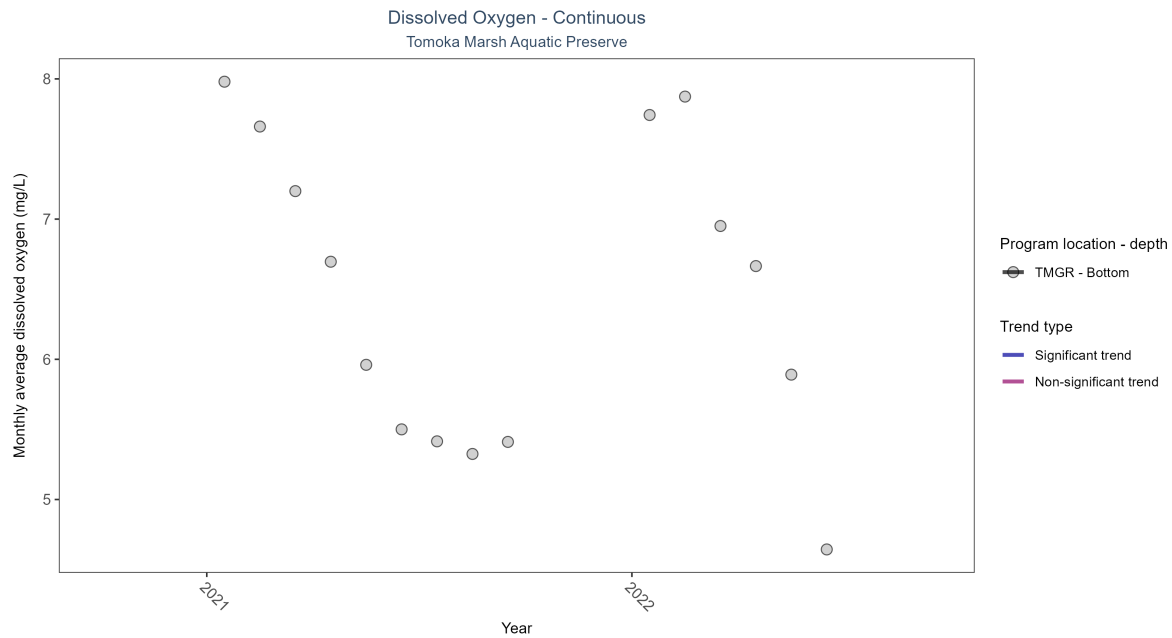


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TMGR	Insufficient data	40492	2	2021 - 2022	6.5	-	-	-	-

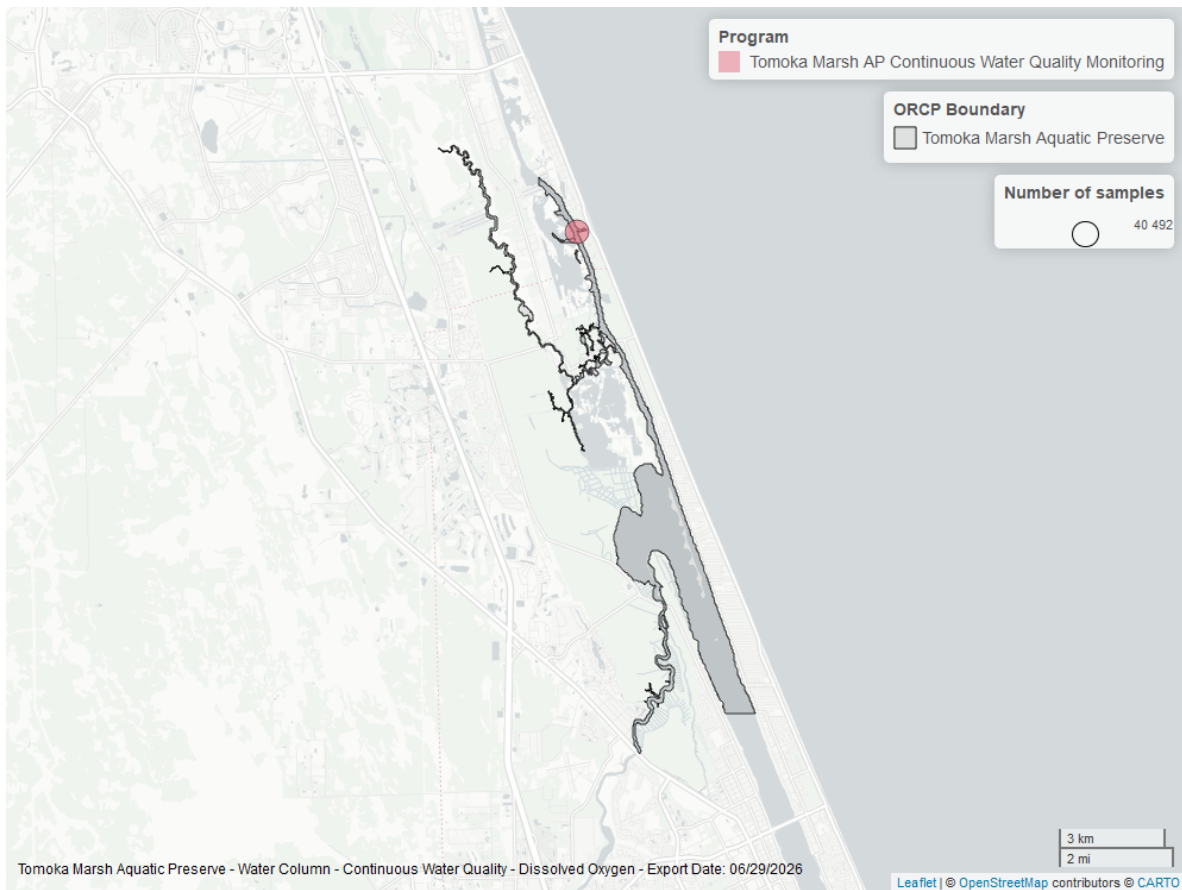


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

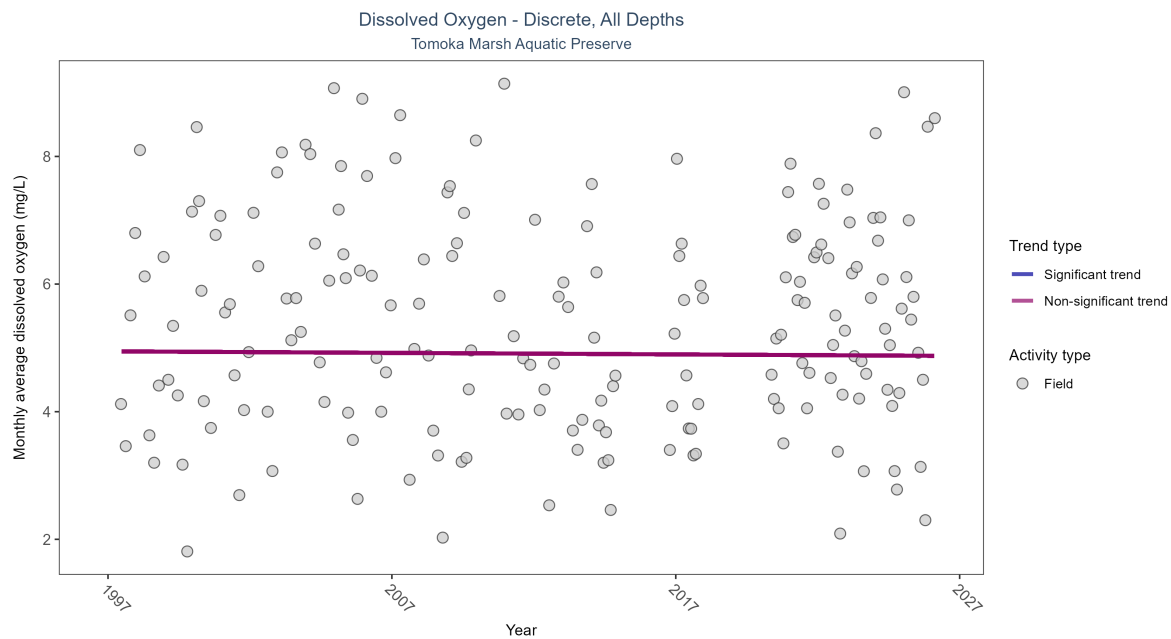


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Dissolved oxygen showed no detectable trend between 1997 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	780	27	1997 - 2026	5.4	0.0132	4.94591	-0.00241	0.6769

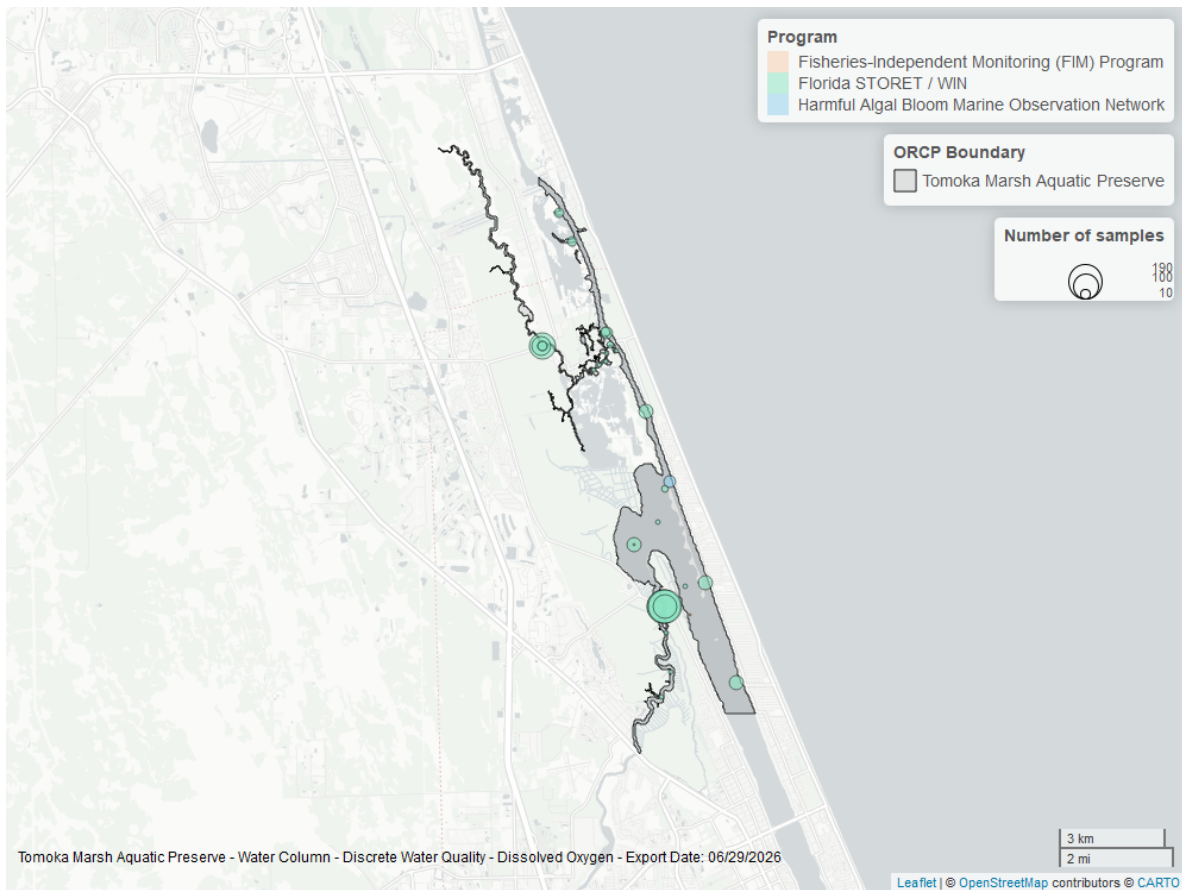


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

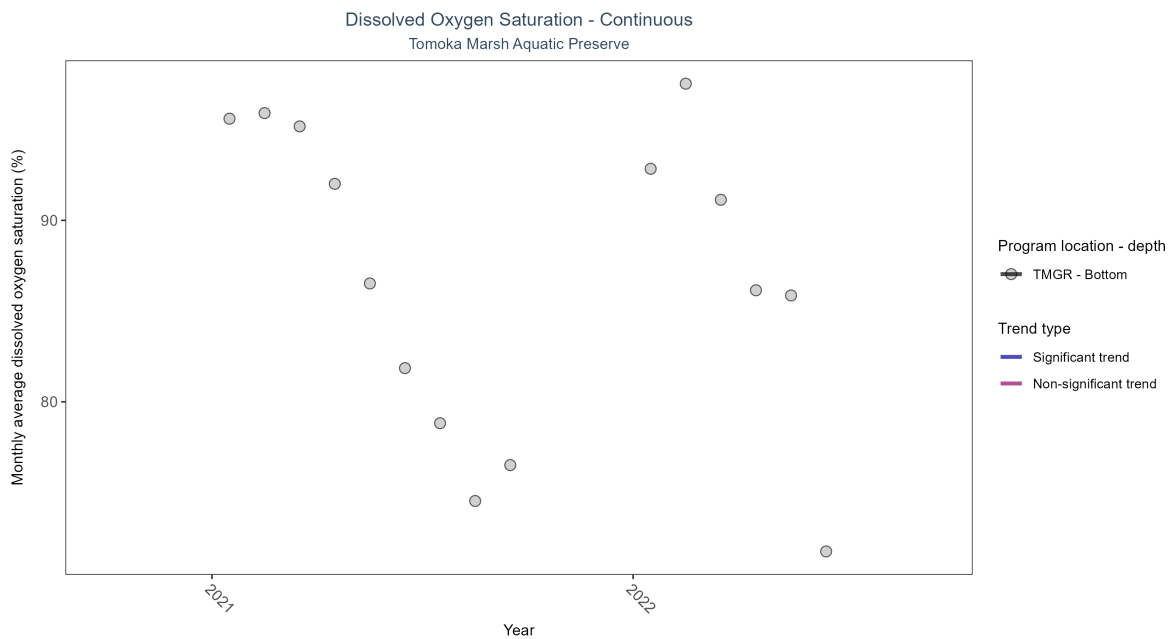


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TMGR	Insufficient data	40492	2	2021 - 2022	88.4	-	-	-	-

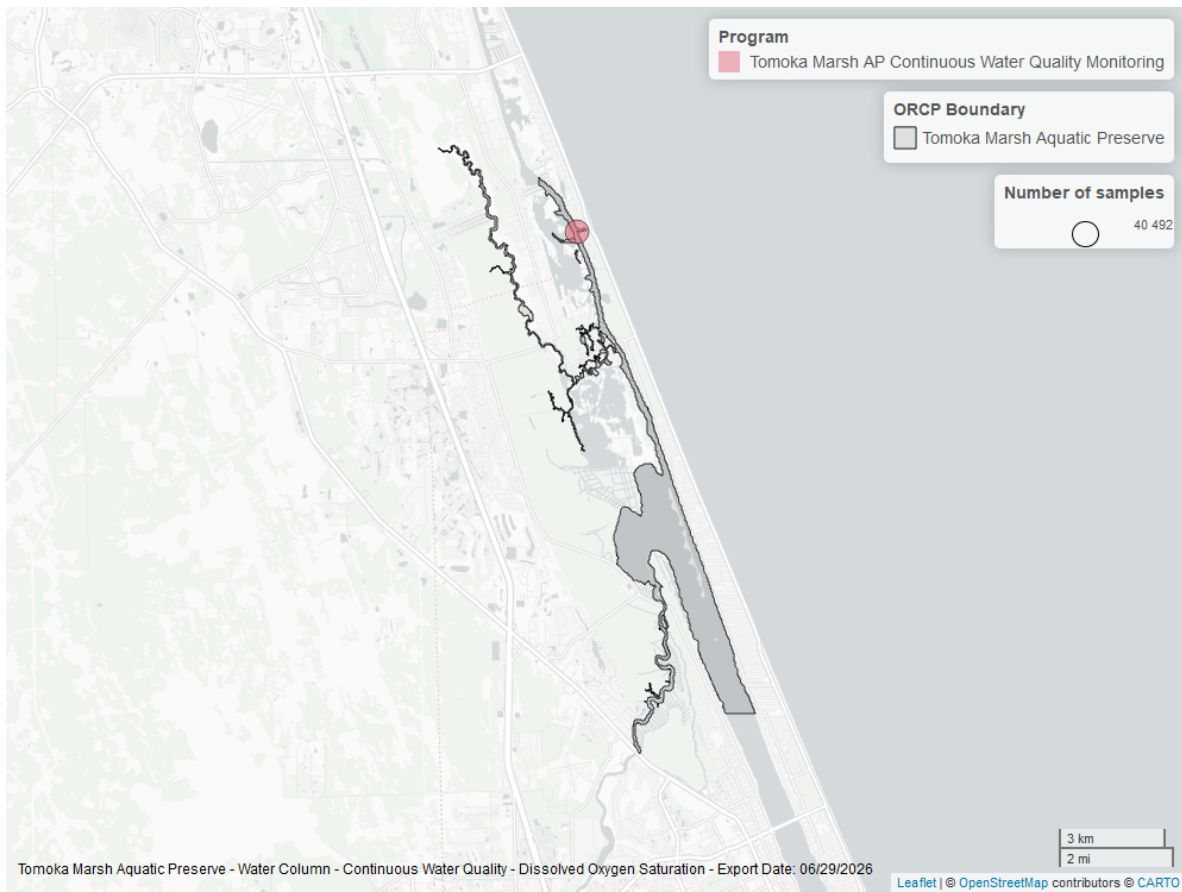


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

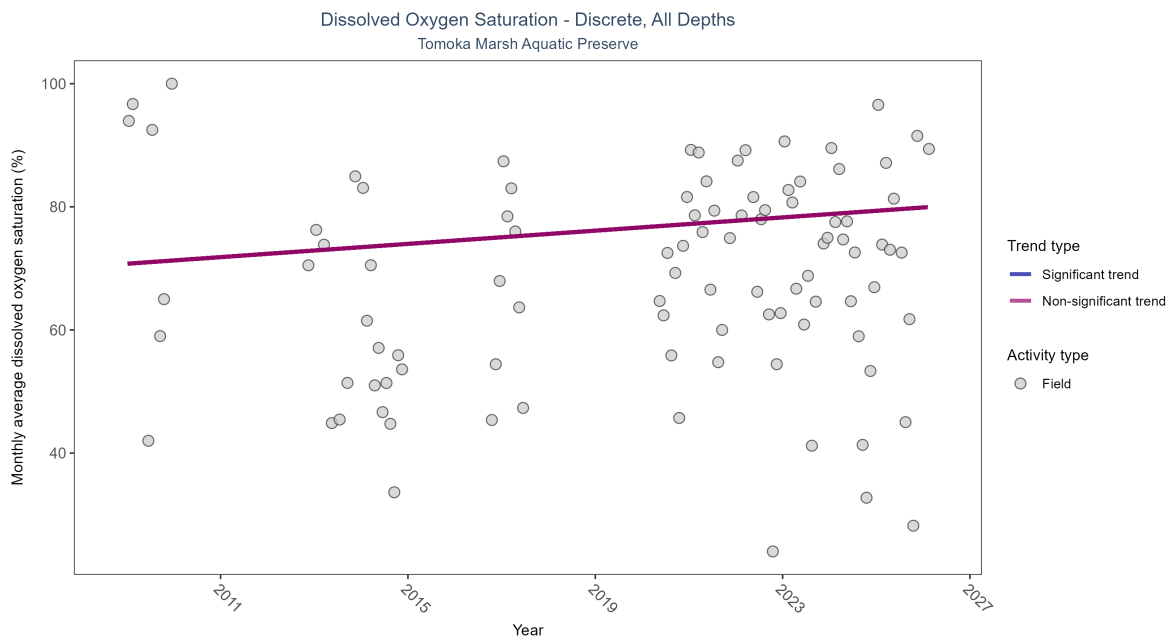


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Dissolved oxygen saturation showed no detectable trend between 2009 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	485	13	2009 - 2026	71.1	0.13139	70.75441	0.5375	0.0563

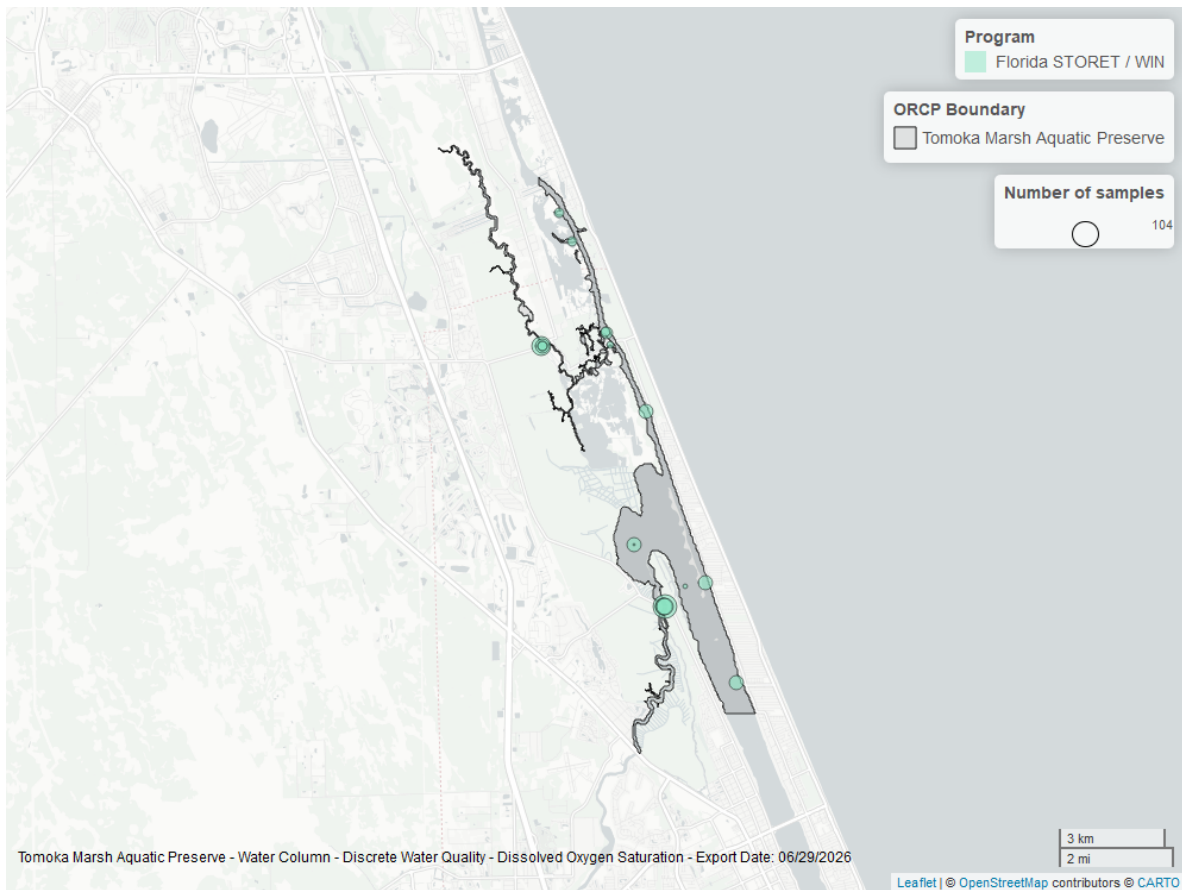


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

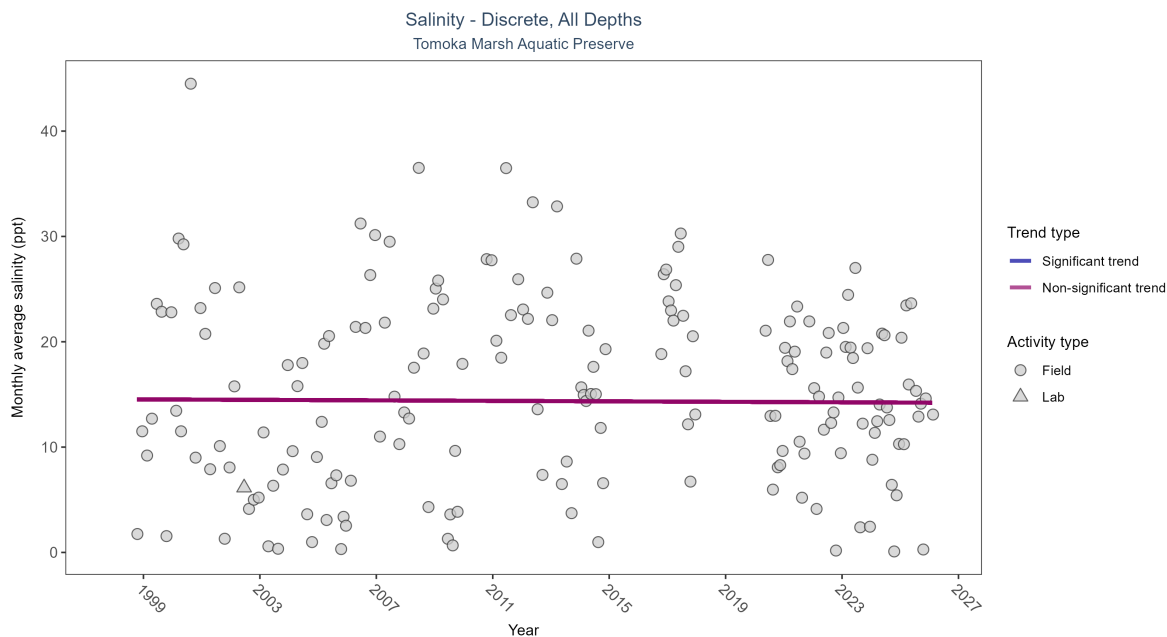


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Salinity showed no detectable trend between 1998 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	773	26	1998 - 2026	15.85	-0.01823	14.54189	-0.01156	0.8385

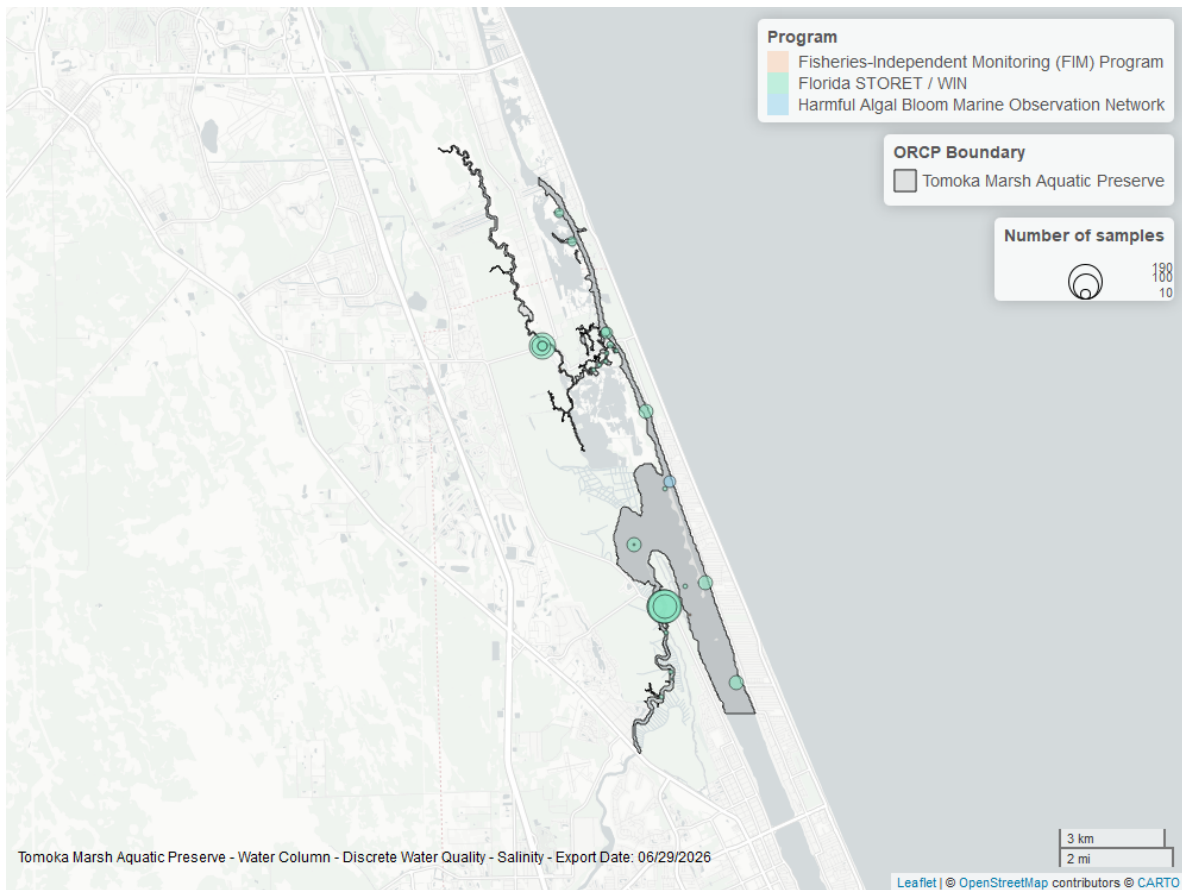


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous

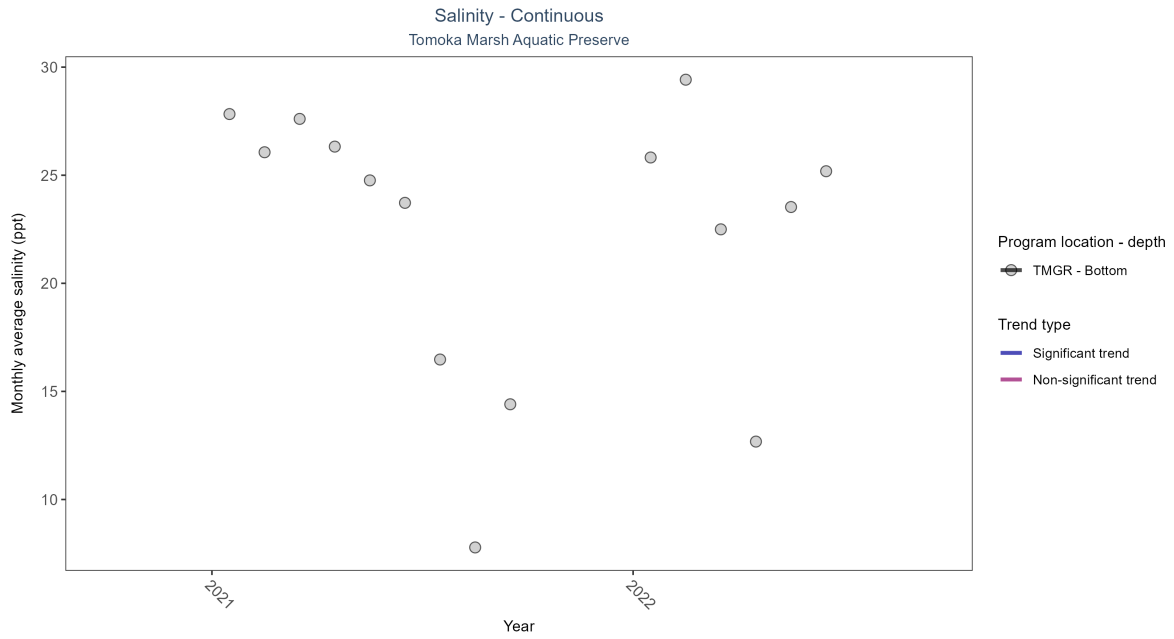


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TMGR	Insufficient data	41374	2	2021 - 2022	23.3	-	-	-	-

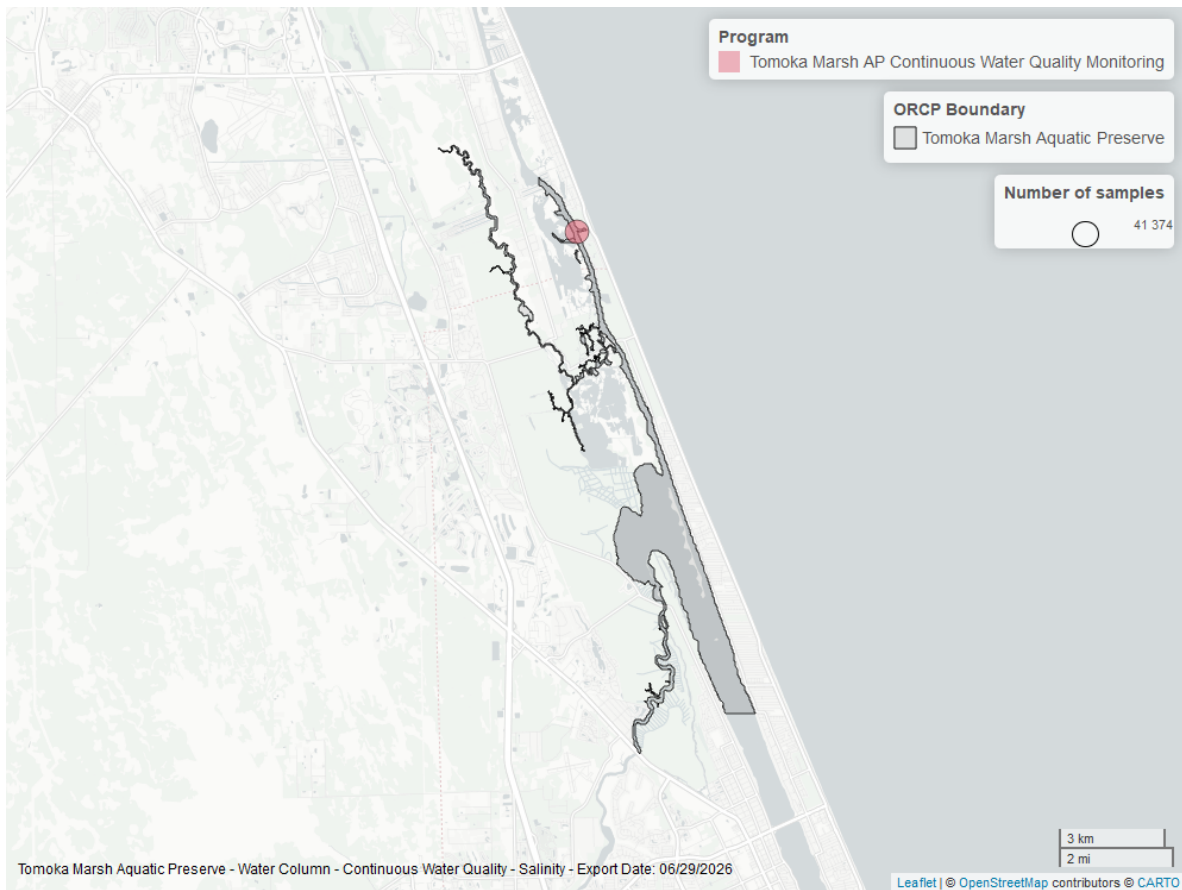


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

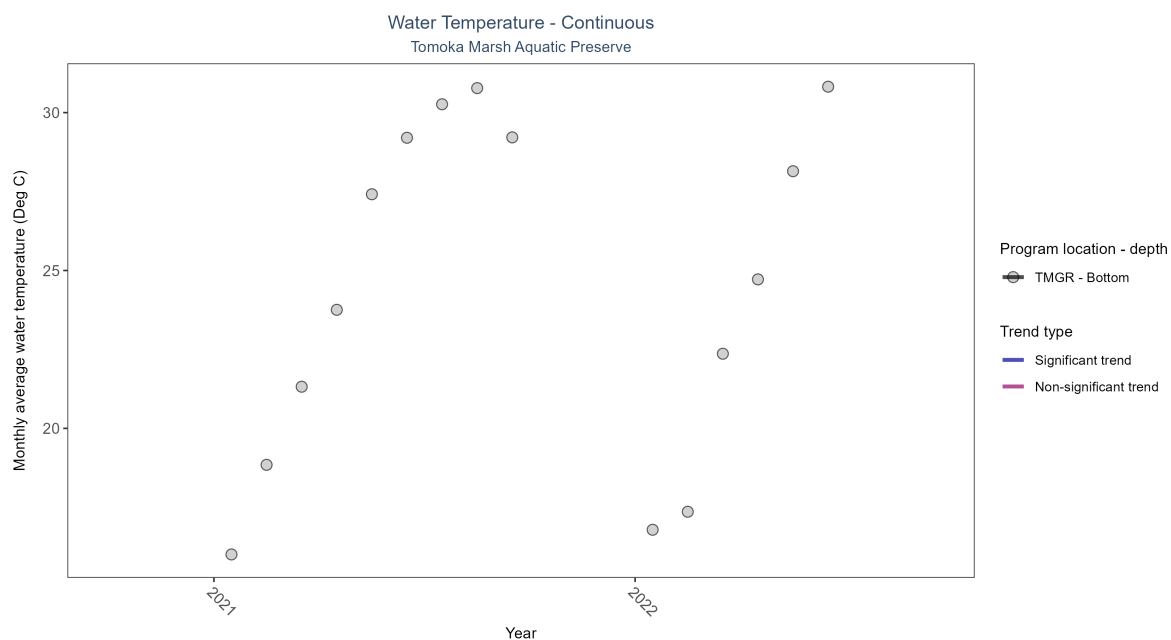


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TMGR	Insufficient data	41374	2	2021 - 2022	25.5	-	-	-	-

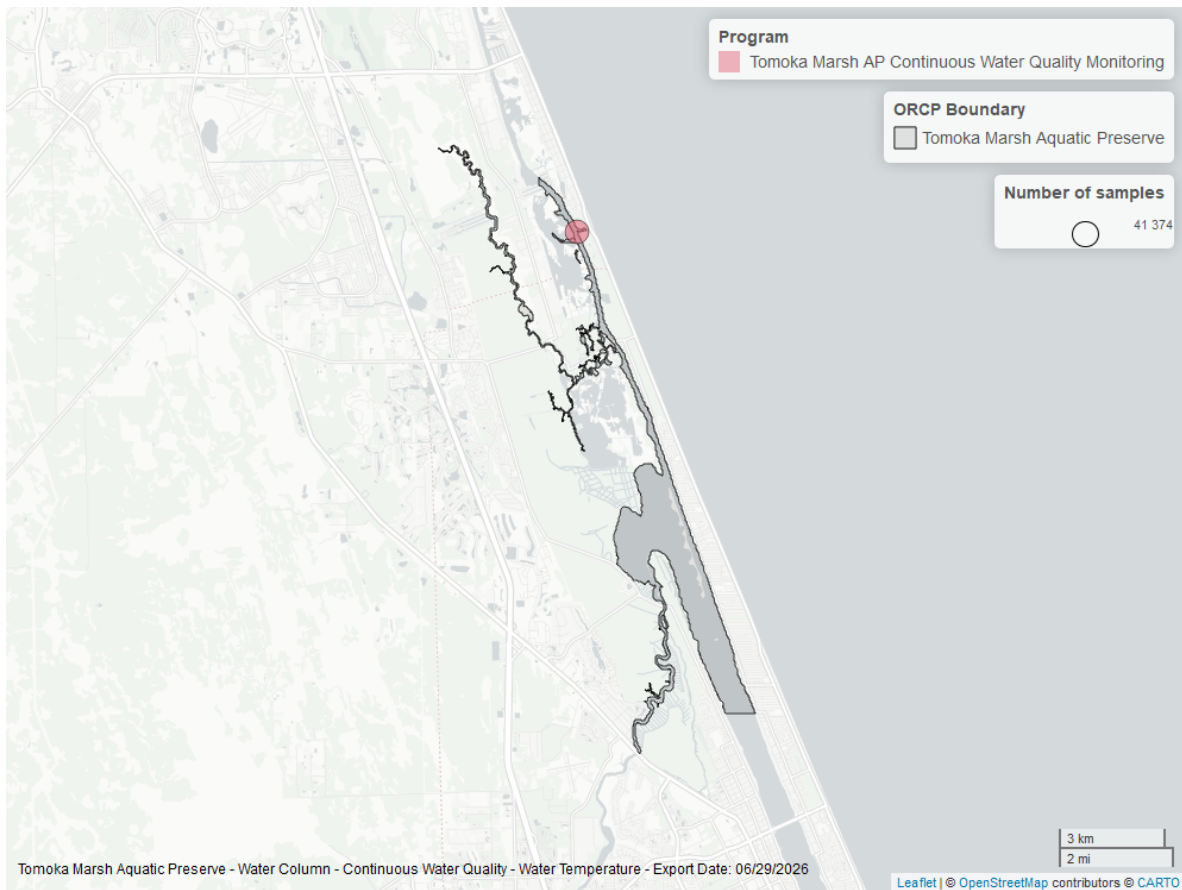


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

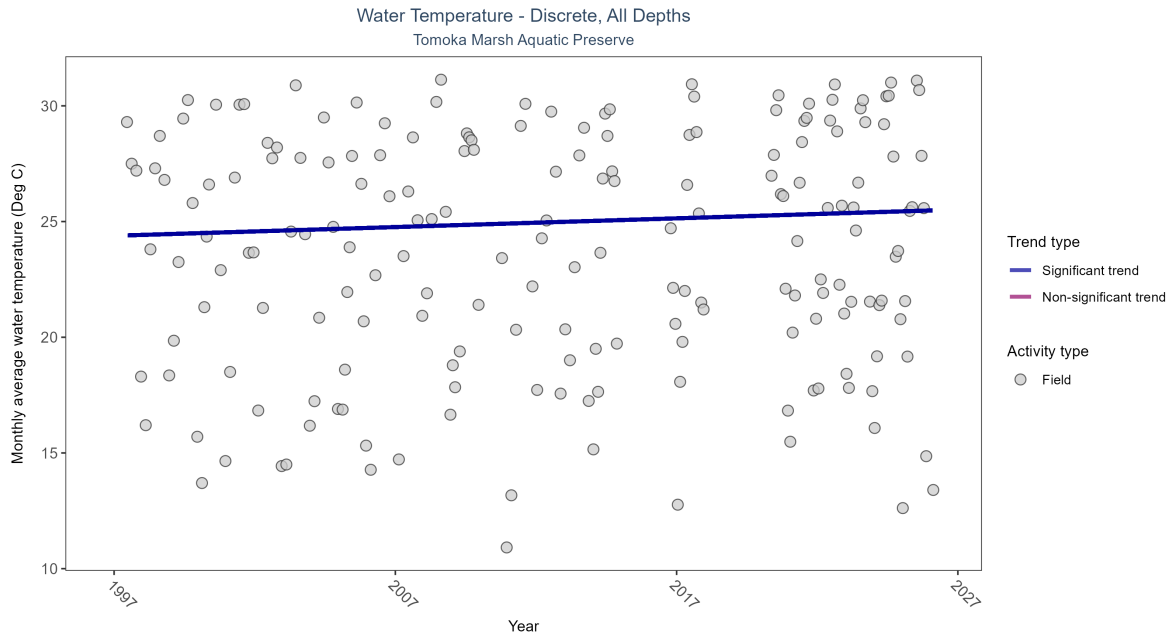


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Monthly average water temperature increased by 0.04Deg C per year.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Increasing trend	796	27	1997 - 2026	25.025	0.12765	24.38472	0.03768	0.0173

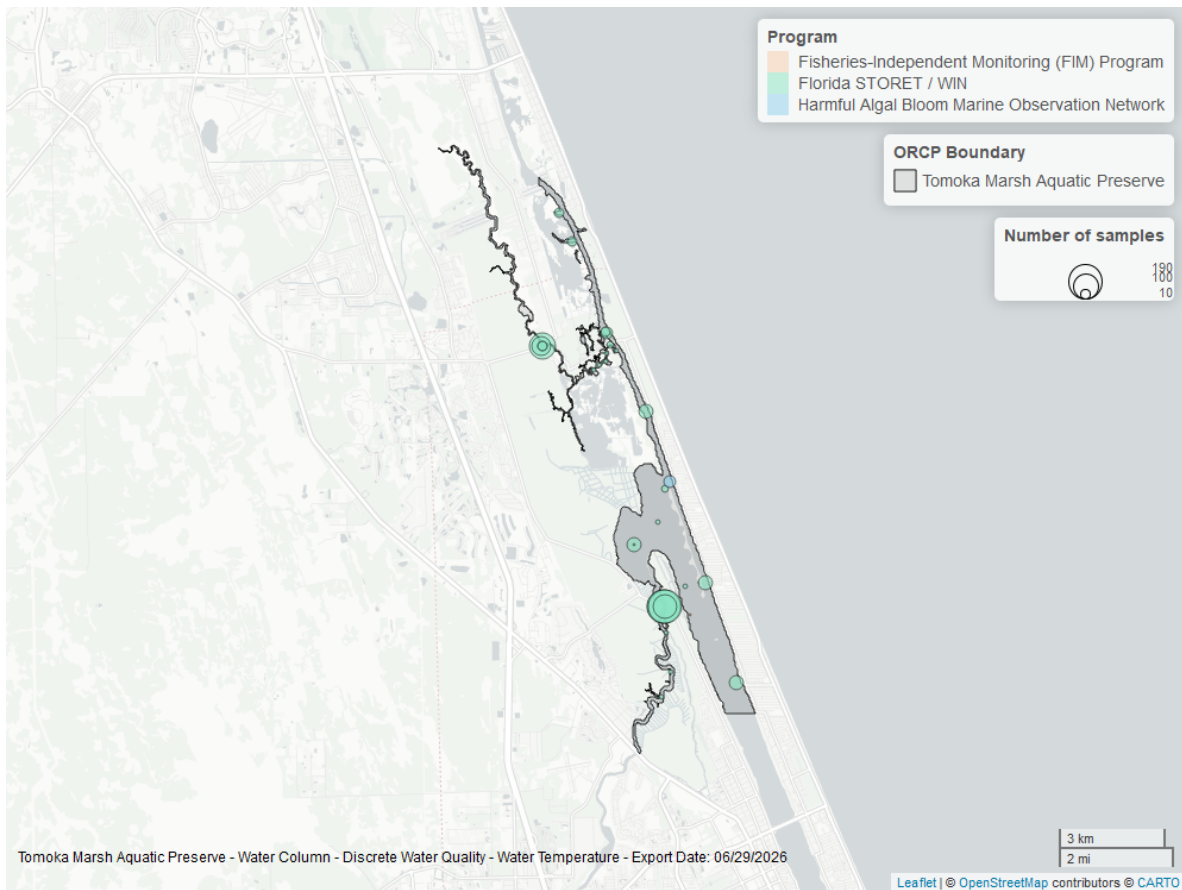


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

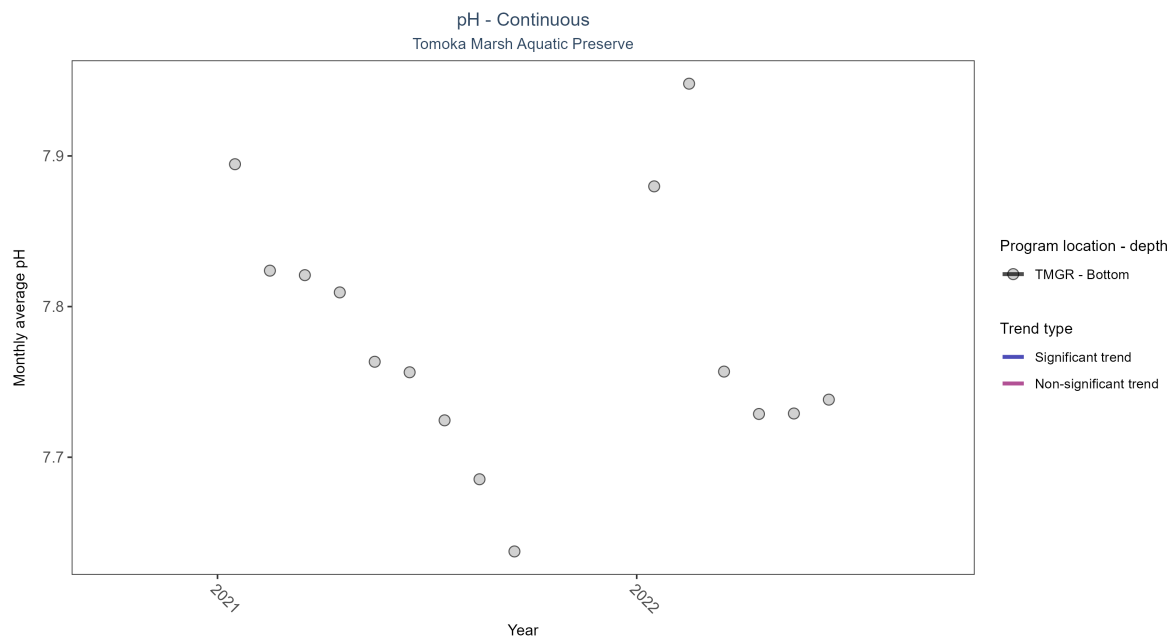


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TMGR	Insufficient data	41374	2	2021 - 2022	7.8	-	-	-	-

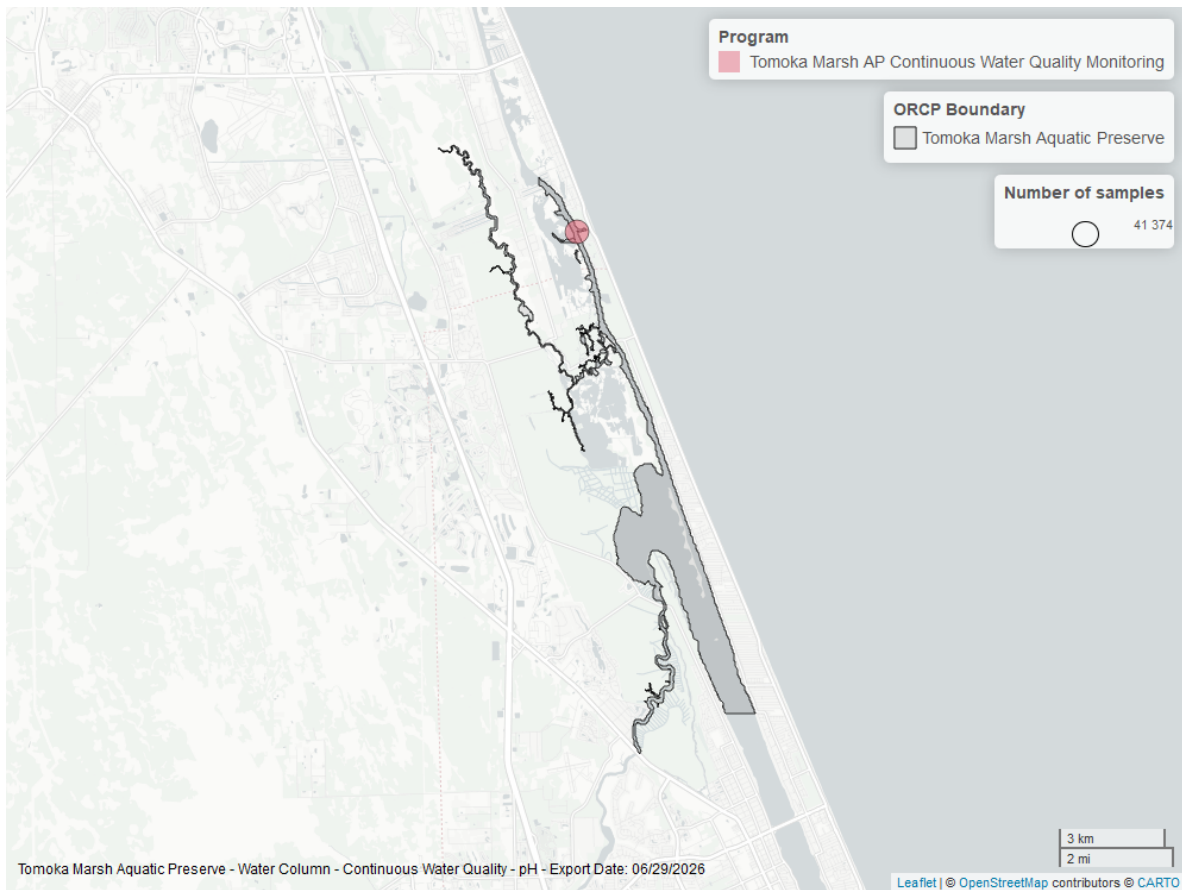


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

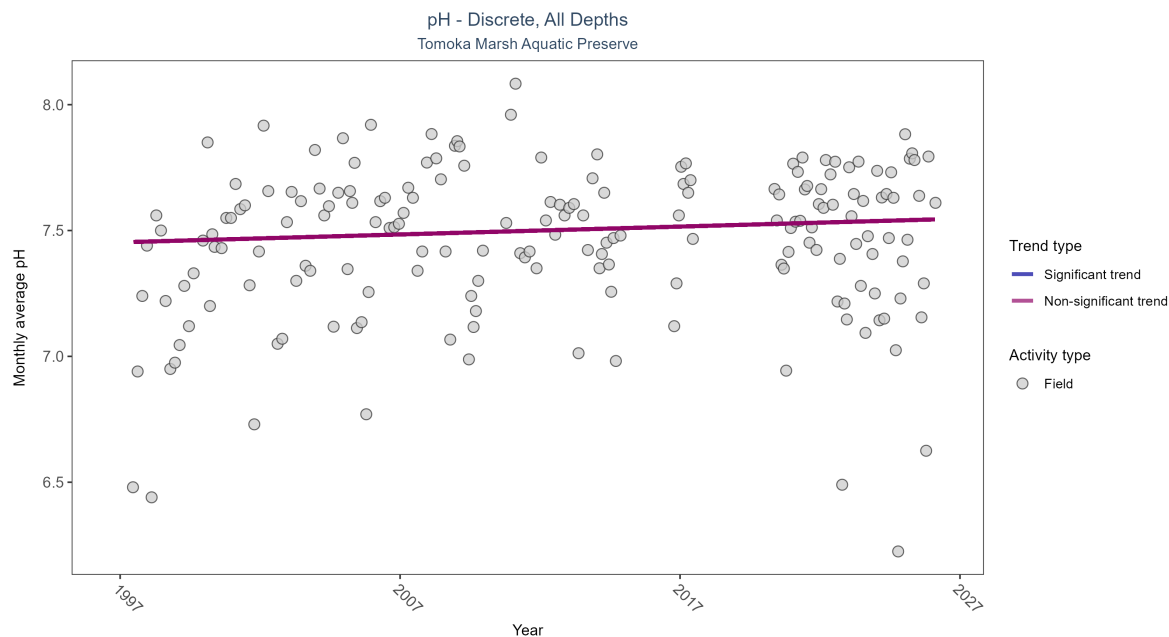


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: pH showed no detectable trend between 1997 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	769	27	1997 - 2026	7.53	0.08866	7.45306	0.00313	0.208

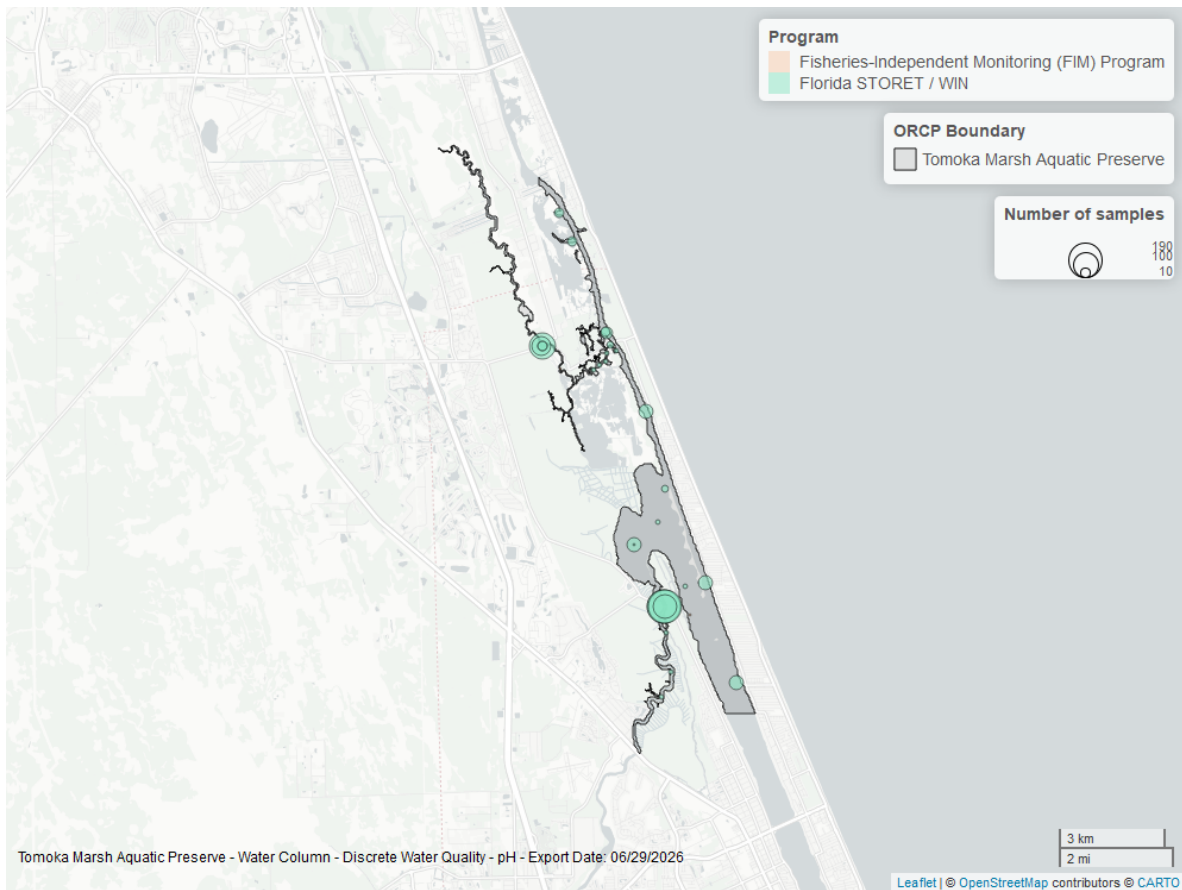


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

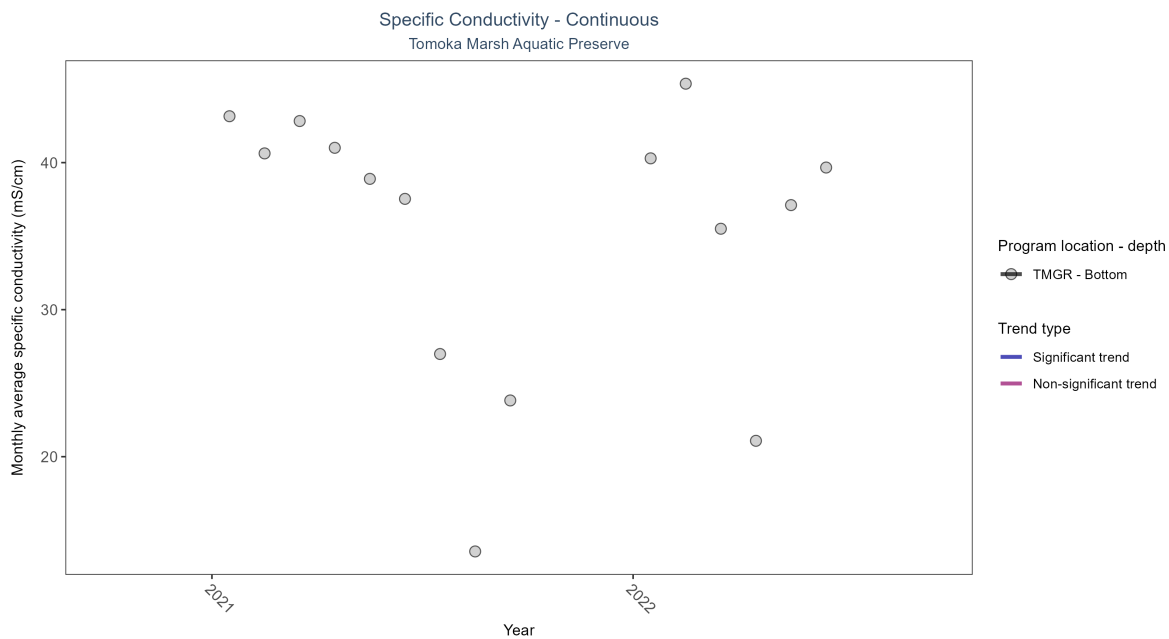


Figure 25: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 13: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TMGR	Insufficient data	41374	2	2021 - 2022	36.84	-	-	-	-

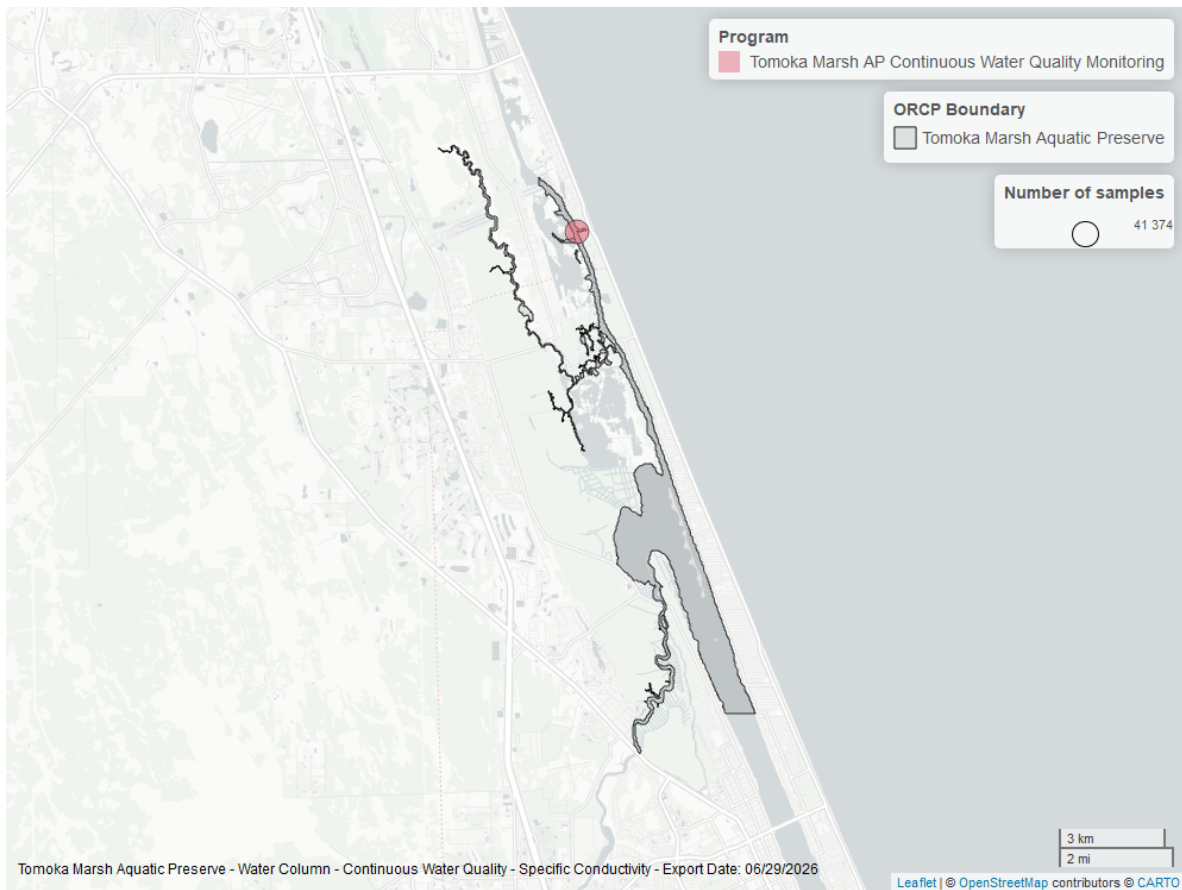


Figure 26: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

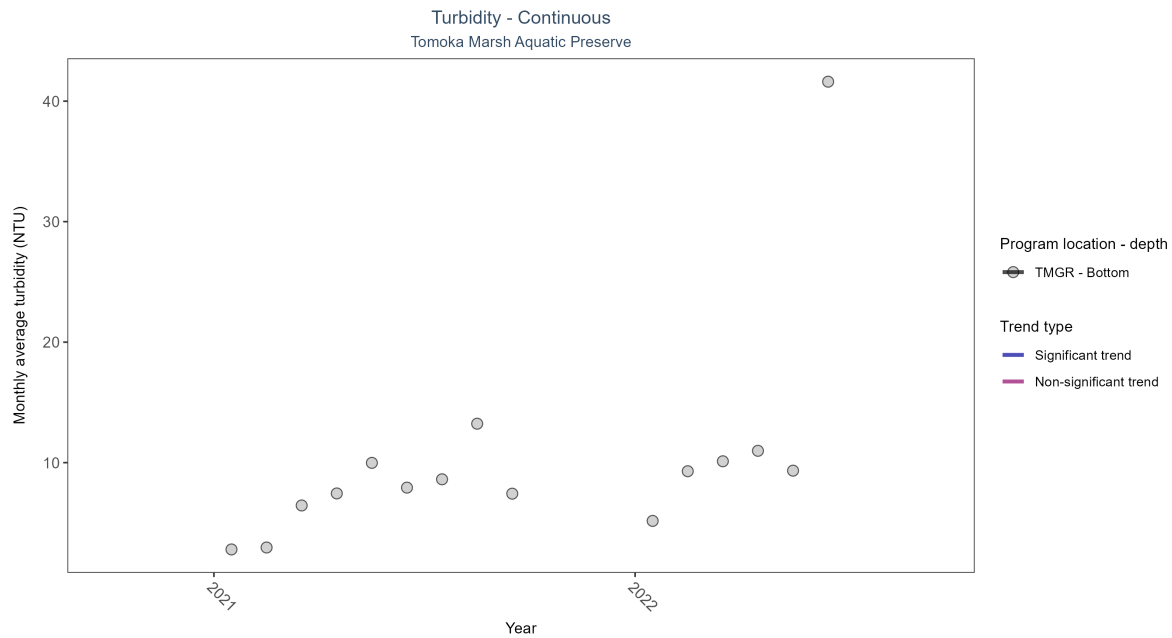


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: There was insufficient data to fit a model for one location.

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
TMGR	Insufficient data	41169	2	2021 - 2022	7	-	-	-	-

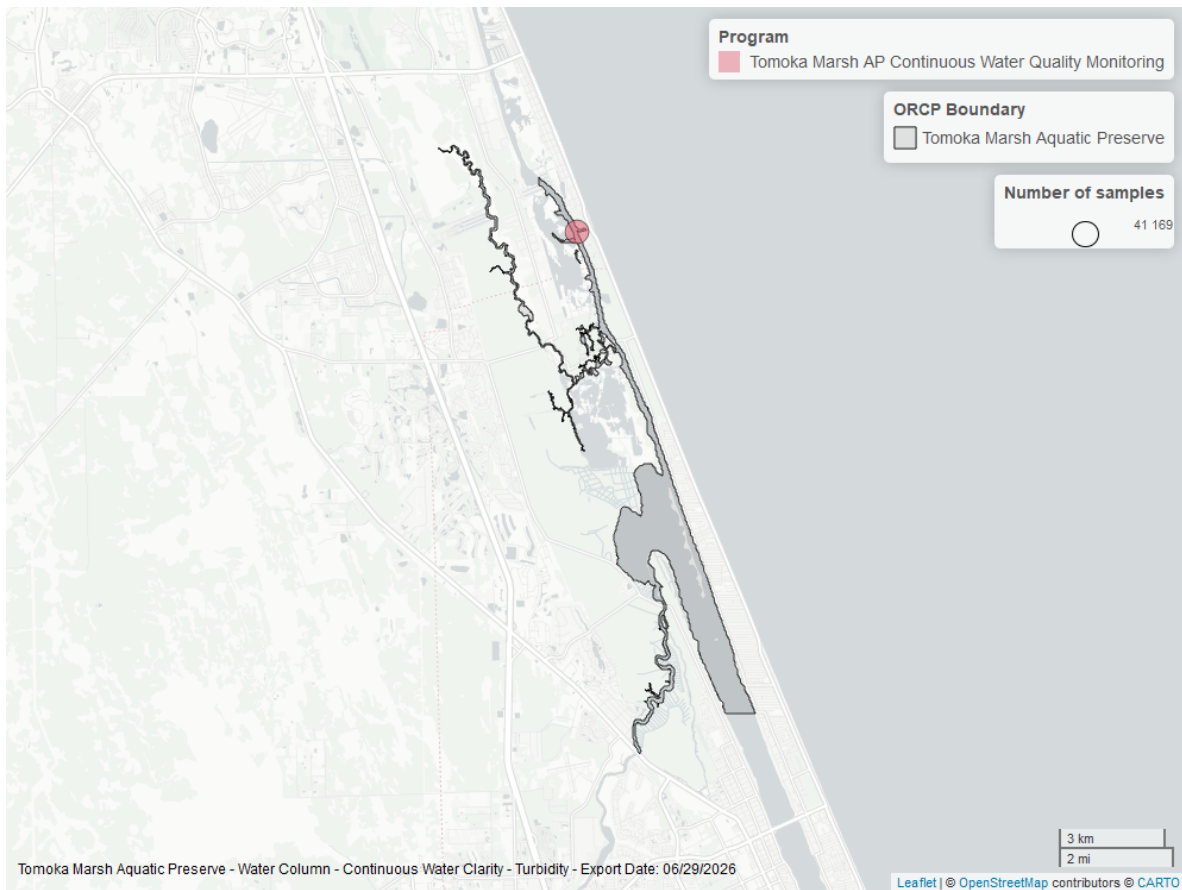


Figure 28: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

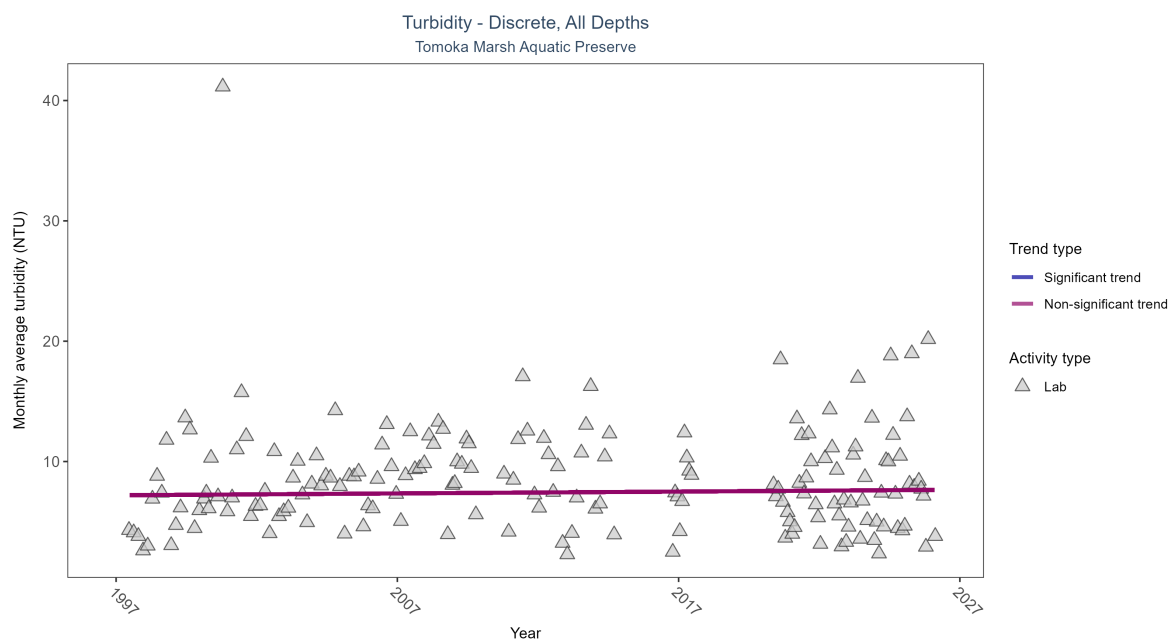


Figure 29: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Turbidity showed no detectable trend between 1997 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	548	27	1997 - 2026	7.27899	0.02767	7.19532	0.01533	0.6138

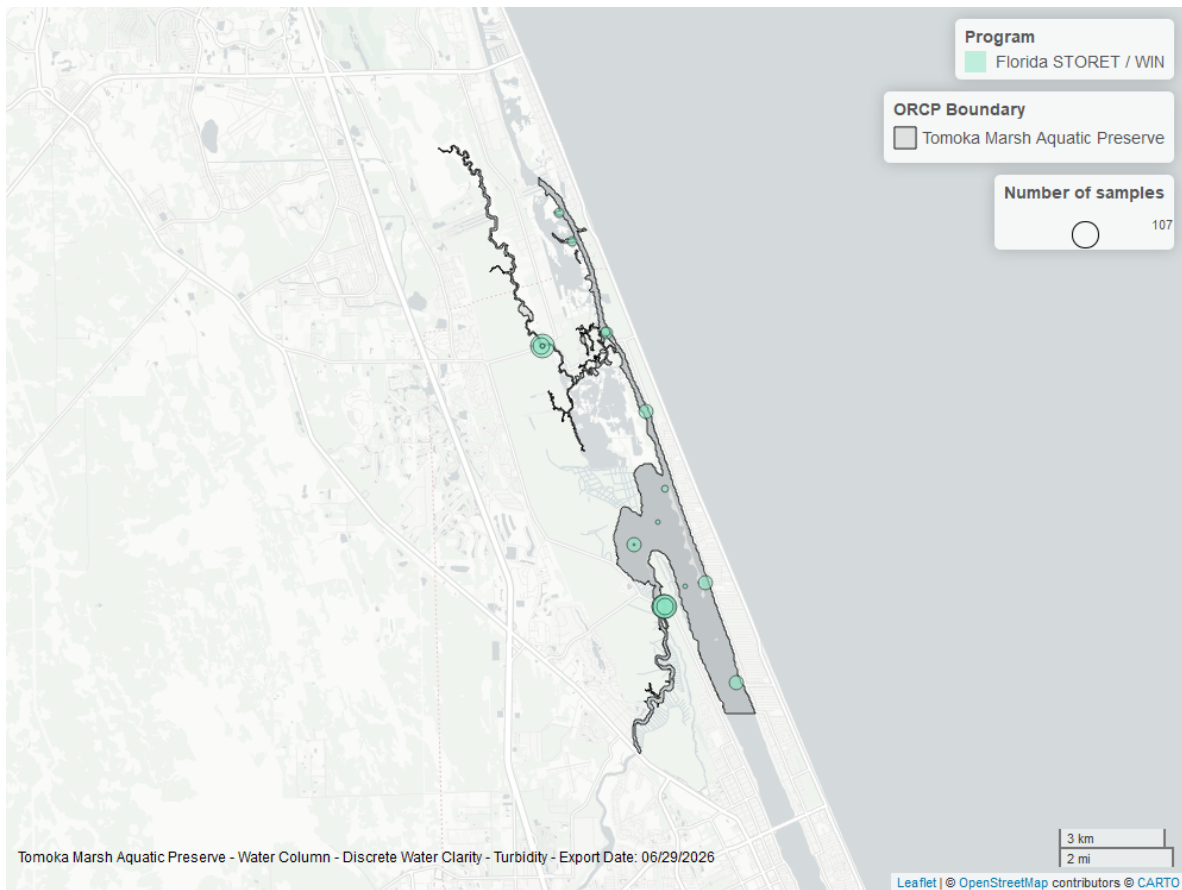


Figure 30: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

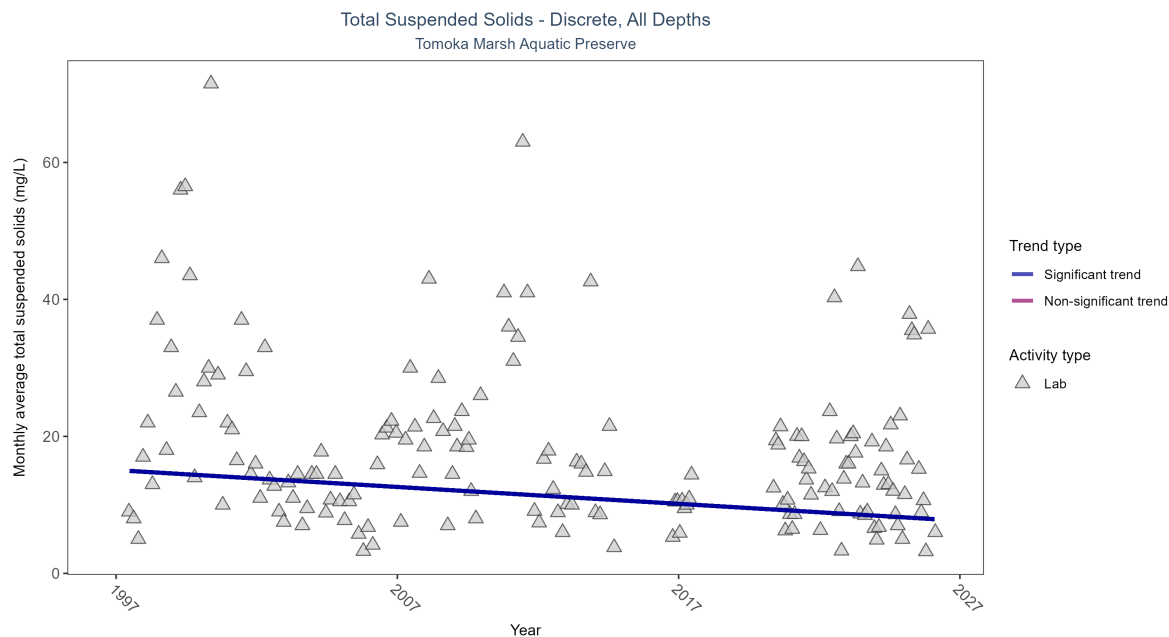


Figure 31: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Monthly average total suspended solids decreased by 0.25 mg/L per year, indicating an increase in water clarity.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Decreasing trend	501	27	1997 - 2026	14	-0.09796	15.07741	-0.24624	0.0085

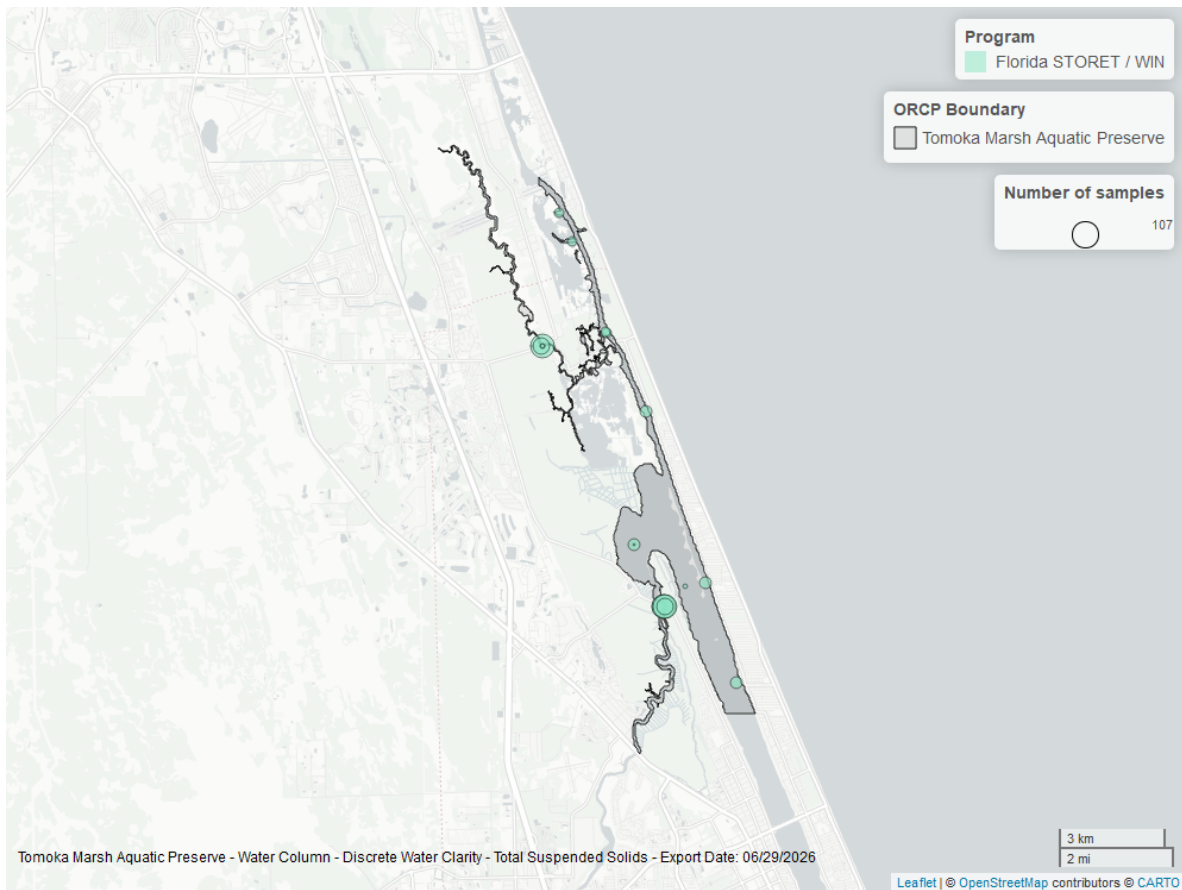


Figure 32: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

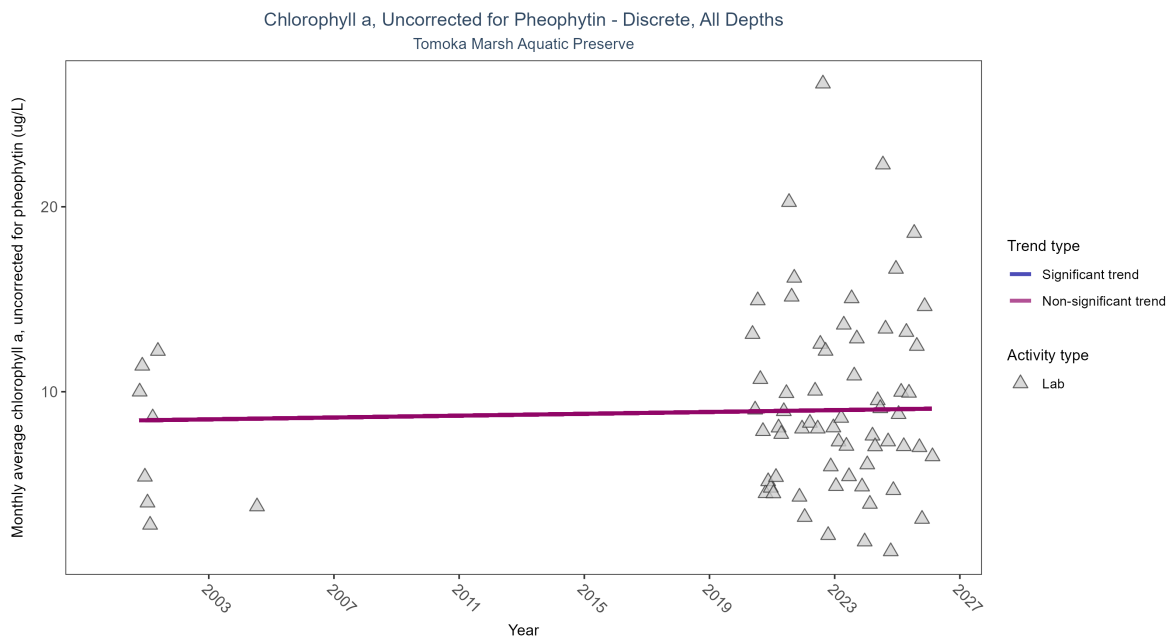


Figure 33: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Chlorophyll a, uncorrected for pheophytin, showed no detectable trend between 2000 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	366	10	2000 - 2026	8	0.00282	8.43173	0.02493	0.9142

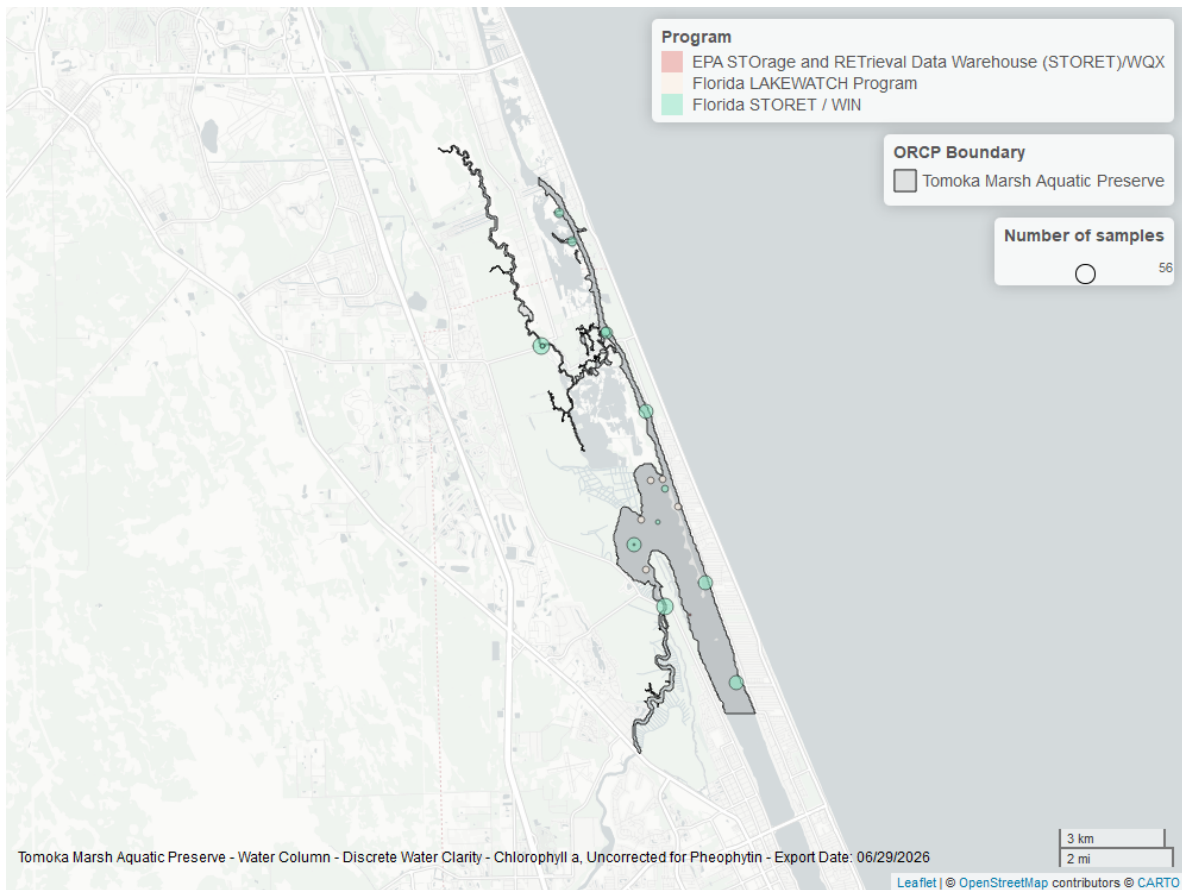


Figure 34: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

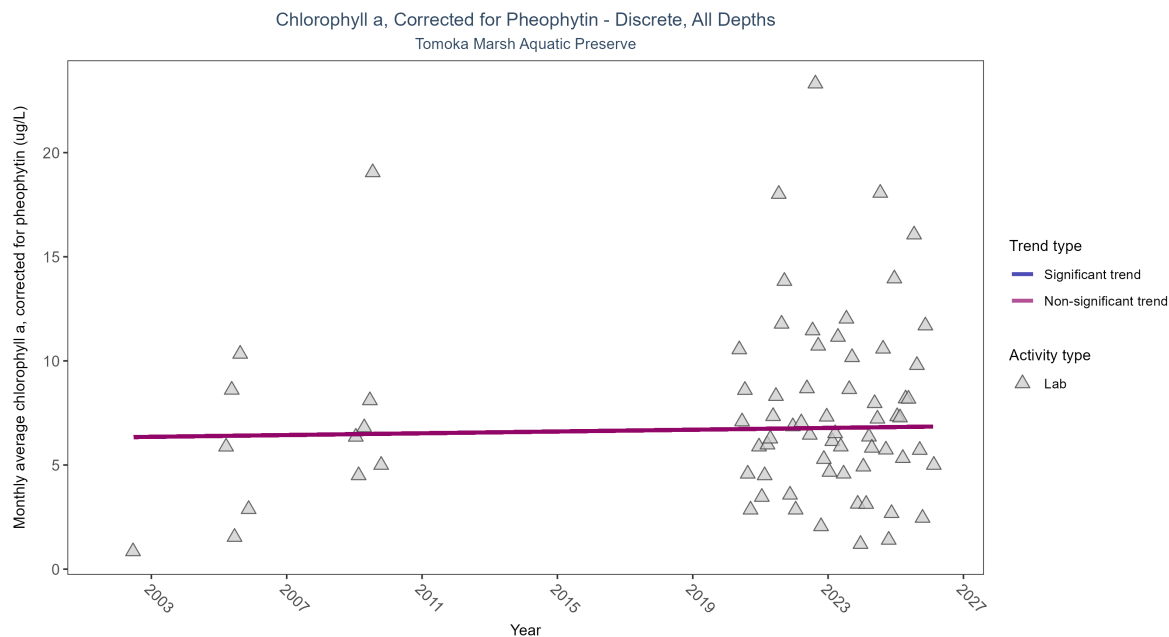


Figure 35: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	339	10	2002 - 2026	6.408	0.05414	6.33176	0.02137	0.5736

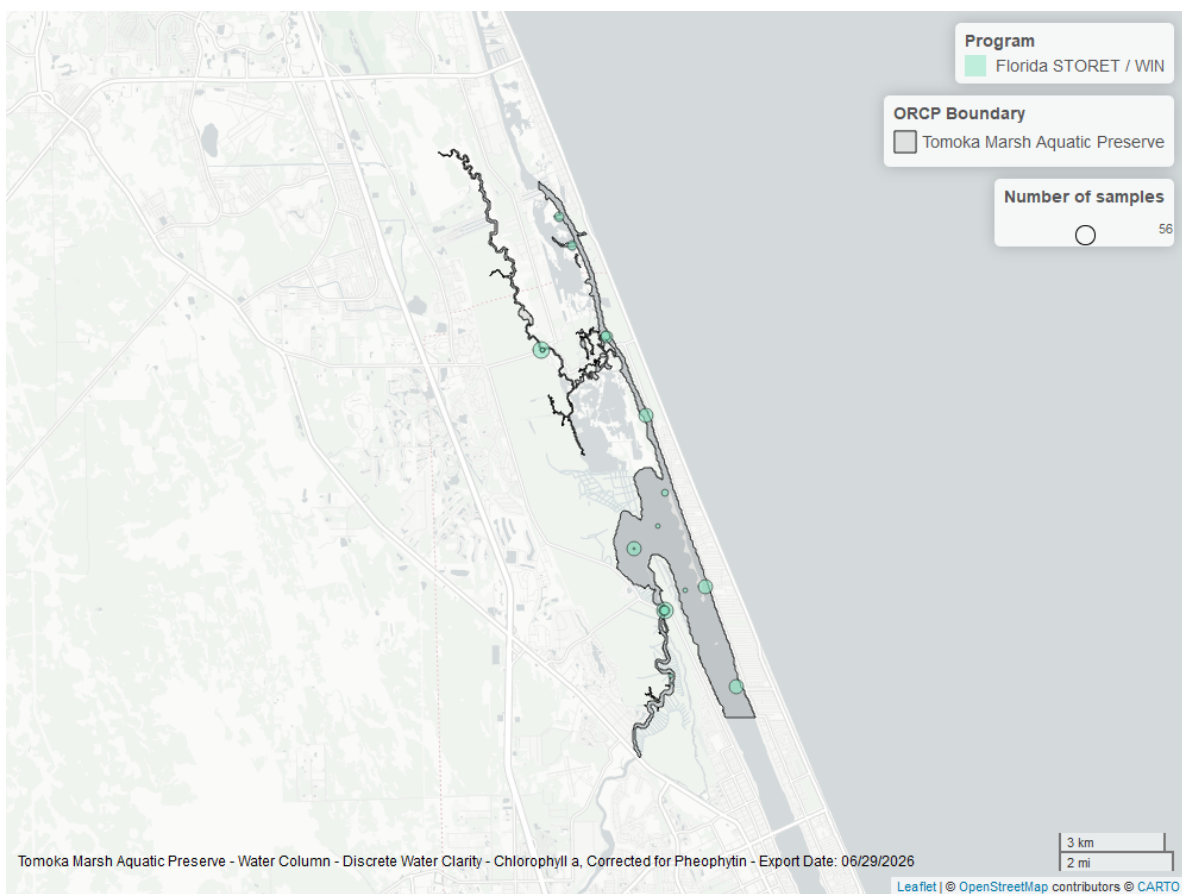


Figure 36: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

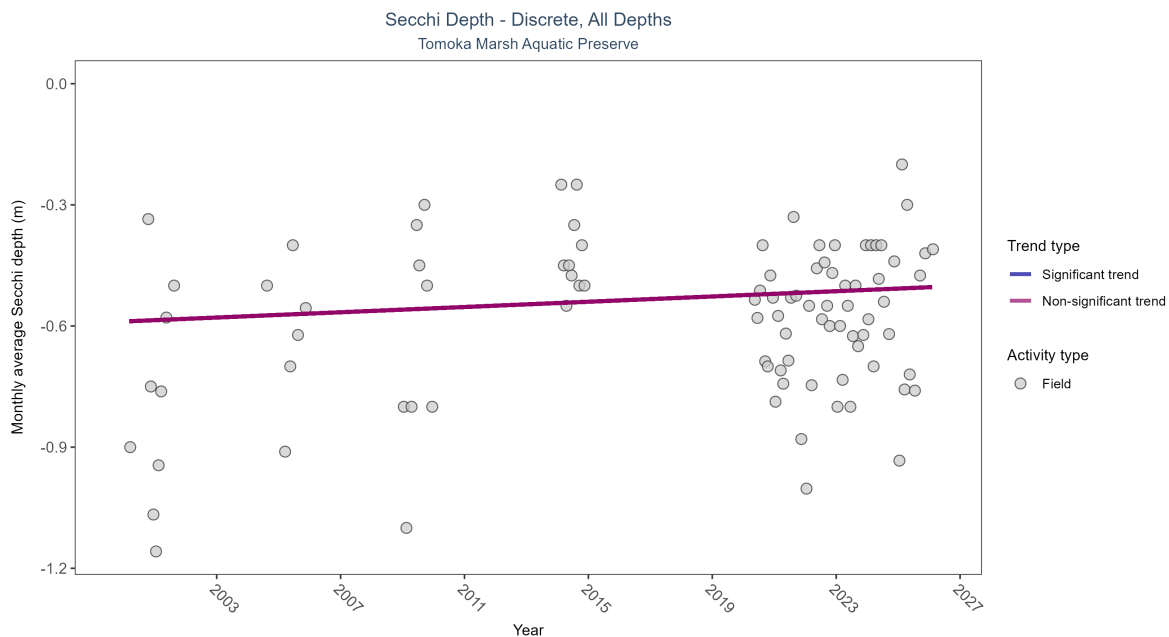


Figure 37: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Secchi depth showed no detectable trend between 2000 and 2026.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	393	13	2000 - 2026	-0.6	0.10973	-0.58883	0.00326	0.2499

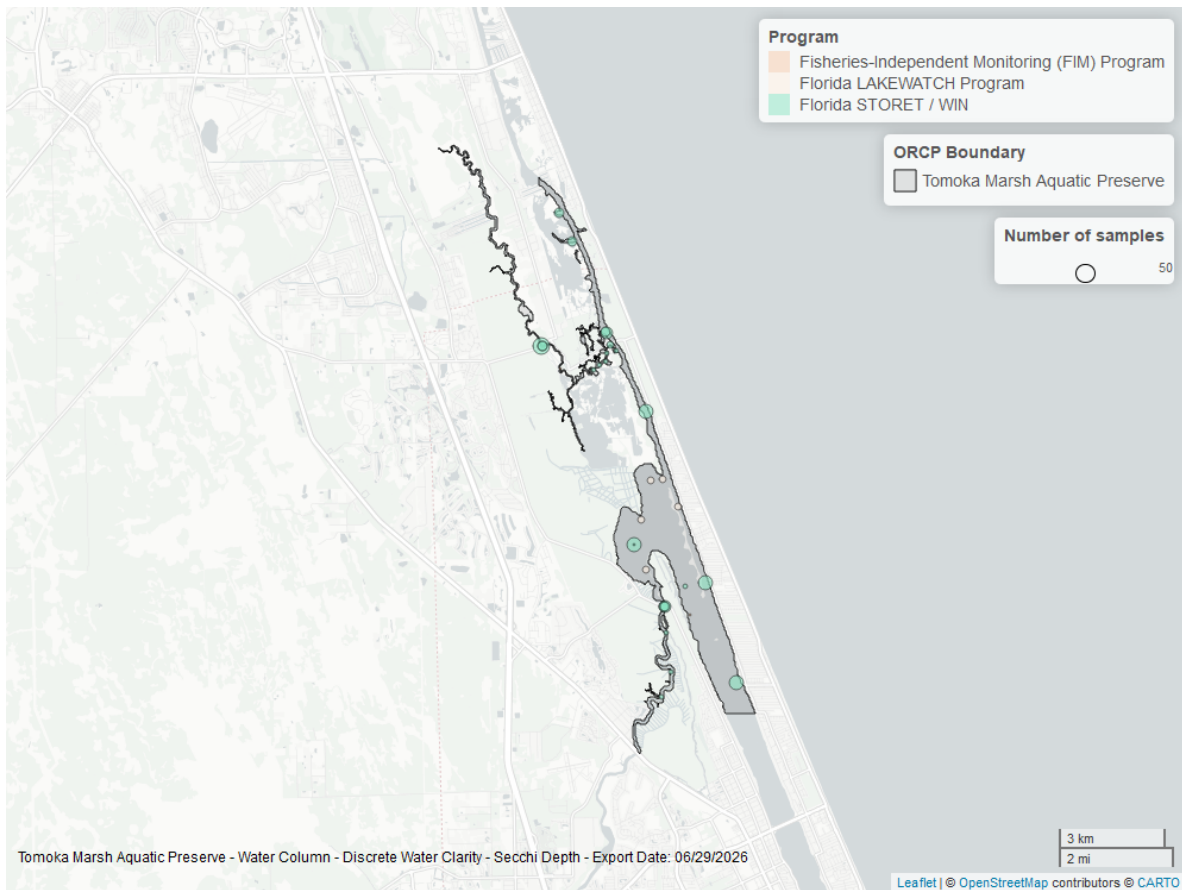


Figure 38: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

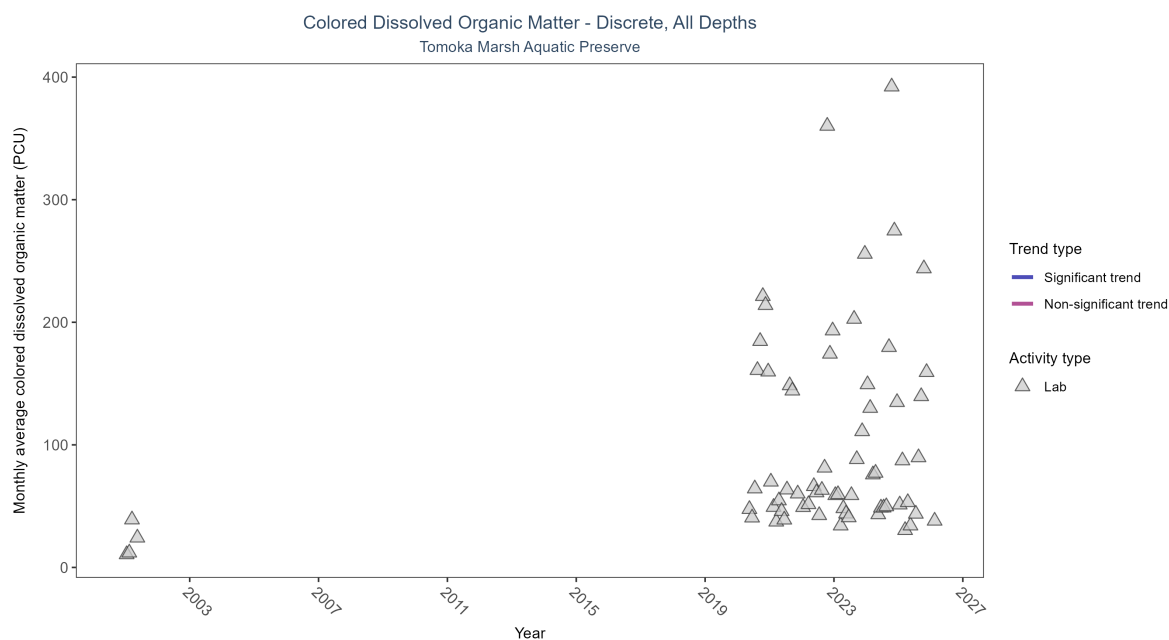


Figure 39: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: There was insufficient data to fit a model for colored dissolved organic matter.

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Insufficient data	328	8	2001 - 2026	58.35273	-	-	-	-

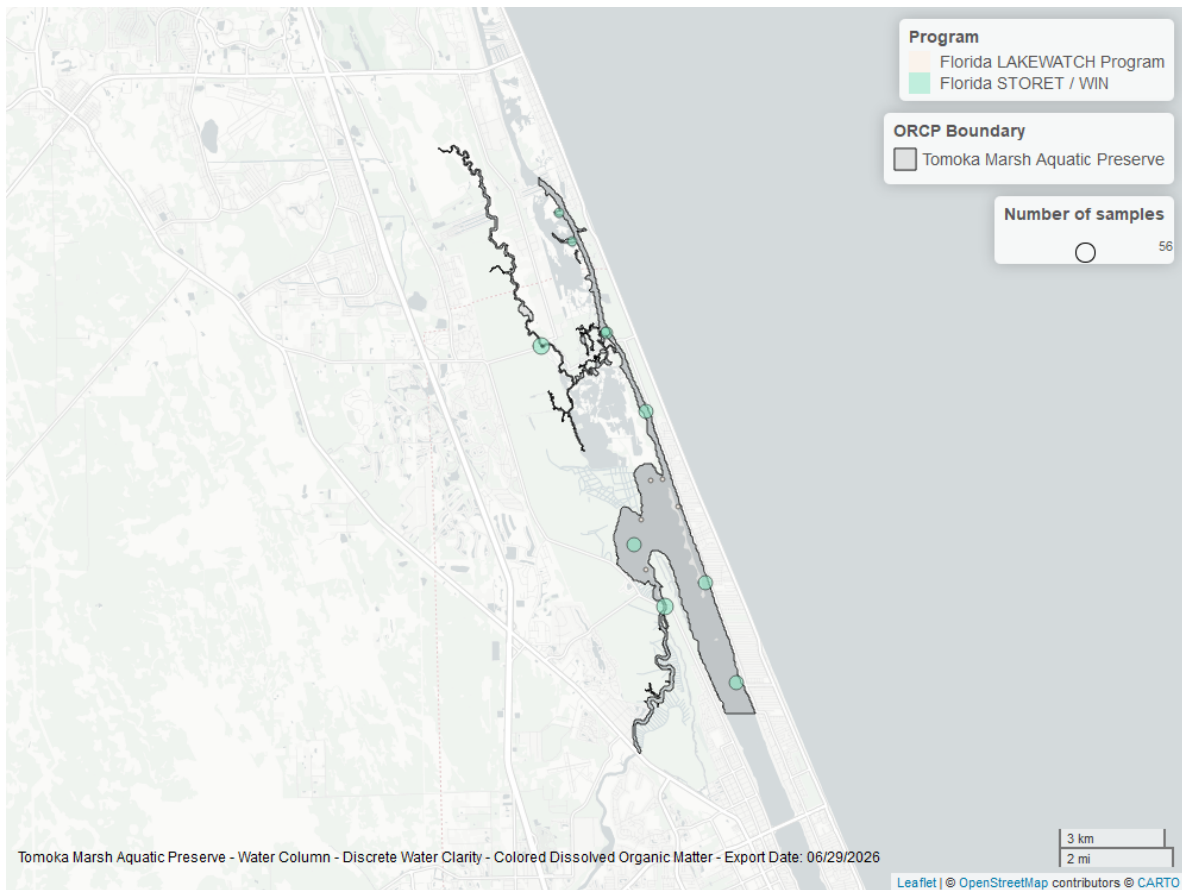


Figure 40: Map showing location of discrete water quality sampling locations within the boundaries of *Tomoka Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.